CORE MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Similarity: properties of similar figures and identifying similar objects from the environment – Centre of Enlargement: establishing the centre using an object and its image – Linear Scale Factor (L.S.F): determining L.S.F from corresponding side lengths of similar figures – Constructing images under enlargement on a plane surface given centre and L.S.F – Enlargement on the Cartesian plane given centre and linear scale factor – Area Scale Factor (A.S.F): working out the ratio of areas of similar plane figures – Volume Scale Factor (V.S.F): working out the ratio of volumes of similar solids – Relationship between L.S.F, A.S.F and V.S.F – Application of similarity and enlargement to real-life situations – Project: making models of similar solids (cubes, cuboids, cylinders, spheres) using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the centre of enlargement and the linear scale factor for similar figures, b) construct the image of an object under an enlargement given the centre and the linear scale factor, c) determine the area and volume scale factor of different figures and solids, d) relate linear scale factor, area scale factor, and volume scale factor in enlargements, e) apply similarity and enlargement to real-life situations, f) appreciate the use of similarity and enlargement in real-life situations.
Core Competencies	– Communication and Collaboration: the learner discusses and recognises similar objects from their immediate environment, sharing observations in group activities and presenting model-building findings to peers. – Creativity and Imagination: the learner makes models of solids of different sizes (cubes, cuboids, cylinders, spheres) using locally available materials, applying similarity and enlargement concepts to design and construct them accurately.
Core Values	– Responsibility: the learner collects and takes care of objects from the immediate environment to recognise those that are similar and discusses what makes them similar. – Integrity: the learner makes models with accurate measurements, adhering to the concept of similarity and enlargement.
Science & Engineering Practices	– Using Mathematics and Computational Thinking: learners compute linear scale factors by taking ratios of corresponding side lengths, derive area and volume scale factors algebraically, and use these relationships to solve quantitative real-life

	problems. – Analysing and Interpreting Data: learners measure corresponding sides of similar figures and solids, organise the data in tables, and identify patterns linking L.S.F, A.S.F, and V.S.F. – Constructing Explanations and Designing Solutions: learners construct geometric images on plain paper and on the Cartesian plane, explain why the construction produces a similar figure, and design a set of scale models as a project solution. – Obtaining, Evaluating, and Communicating Information: learners use digital devices and other resources to explore real-life applications of similarity and enlargement (maps, architectural plans, Kenyan road signage, National Parks maps), then communicate findings to the class.
Pertinent & Contemporary Issues (PCIs)	– Social Cohesion: the learner works in a group to create models of various similar solids, building teamwork and respect for peers' contributions across different backgrounds. – Education for Sustainable Development (Environmental Education): the learner collects and uses objects from the immediate environment (e.g., tins, bottles, seed pods found around the school compound) and discusses with peers to recognise those that are similar, promoting environmental awareness and responsible use of local resources.
Career Connections	– Architecture and Construction: architects and building contractors in Kenya use similarity and scale factors when reading and drawing construction plans for houses and public buildings — knowledge of L.S.F, A.S.F, and V.S.F is directly applied when scaling blueprints. – Cartography and Surveying: land surveyors and cartographers at Kenya's Survey of Kenya use enlargement and scale to produce maps — the ratio between map distances and real ground distances is a direct application of linear scale factor. – Engineering and Manufacturing: engineers designing machinery, vehicles, or agricultural equipment use geometric similarity to create prototype models before full-scale production. – Fine Art, Design, and Advertising: graphic designers and artists in Nairobi's creative industry use enlargement principles when resizing logos, posters, and digital artwork while preserving proportions. – Medical Imaging and Pharmacy: radiographers and medical laboratory technicians use scale factors when interpreting X-rays, scans, and microscope images to relate image dimensions to actual anatomical sizes.
Focus for Lessons	This unit develops learners' understanding of similarity and enlargement as tools for describing how shapes and solids scale in the real world. Beginning from concrete Kenyan contexts — such as the scaled floor plans of Nairobi city buildings, the proportional artwork on Maasai shields, and the similar shapes of maize cobs of different sizes — learners progress from identifying similar figures and their properties, through constructing enlargements geometrically and on the Cartesian plane, to establishing and applying the powerful relationships between linear, area, and volume scale factors. A hands-on model-building project using locally available materials anchors abstract concepts in practical, creative work.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: When you scale up a shape or solid — like doubling the size of a water tank or enlarging a building plan — what exactly happens to its lengths, its area, and its volume, and how can we use this to solve real problems in our community? KICD KEY INQUIRY QUESTION: 1. How are similarity and enlargement applied in day-to-day life?
Anchoring Phenomenon	A Grade 10 class in Kisumu notices that their school's architectural plan pinned to the staffroom wall shows the entire school compound fitting on a single A3 sheet of paper, yet every classroom block, the water tank, and the football pitch are clearly recognisable and perfectly proportional to the real structures. The plan is labelled "Scale 1 : 200." Students are puzzled: How can a building that is 10 metres wide appear as only 5 centimetres on paper — and yet every window, door, and corridor is still in exactly the right proportion? They also notice that two water storage tanks on the compound look identical in shape but one holds exactly 8 times as much water as the other, even though it appears only twice as tall. This sparks a deeper question: if you scale up a shape by a factor of 2 in length, does the area double and the volume double as well — or does something very different happen? The

	phenomenon is anchored in a familiar, tangible Kenyan school environment and generates genuine cognitive conflict between learners' intuitive expectation (double the size → double everything) and the mathematical reality of squared and cubed scale factors.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Properties of Similar Figures: Learners examine the school architectural plan and the two water tanks (anchoring phenomenon). In groups they collect objects from the school environment (tins, exercise books, seed pods) and sort them into similar and non-similar sets. They discuss and review the properties of similar figures — equal corresponding angles, proportional corresponding sides — and record observations on a group chart. The driving question is introduced and posted on the Driving Question Board (DQB). Lesson 2 — Centre of Enlargement and Linear Scale Factor (L.S.F): Using an object drawn on plain paper and its enlarged image, learners draw lines through corresponding vertices and discover that all lines meet at a single point — the centre of enlargement. They measure corresponding sides and compute L.S.F = image length ÷ object length. They connect this back to the school plan: L.S.F = 1/200. Learners update the DQB with new understanding about what L.S.F means. Lesson 3 — Constructing Images under Enlargement on a Plane Surface: Learners are guided to draw the image of a given polygon (e.g., a triangle representing a classroom block) on plain paper under a given centre and L.S.F (positive and negative values). They follow a step-by-step construction process using rulers and a geometrical set. Pairs compare constructions and check accuracy by verifying that all image sides are in the correct ratio to the object sides. They revisit the architectural plan phenomenon to confirm understanding. Lesson 4 — Enlargement on the Cartesian Plane: Learners plot a given object on the Cartesian plane and construct its image under enlargement with a specified centre (including the origin) and L.S.F. They practise with both positive L.S.F (image on same side as object) and negative L.S.F (image on opposite side). Connections are made to GPS grid coordinates used in Kenyan land surveying. The Cartesian plane work is linked to the Kisumu school-plan phenomenon by treating the plan as a coordinate grid. Lesson 5 — Area Scale Factor (A.S.F): Learners draw pairs of similar plane figures (rectangles, triangles) with known L.S.F, calculate their areas, and tabulate L.S.F vs. ratio of areas. They discover A.S.F = $(\text{L.S.F})^2$. They apply this to the Unga Pembe flour packet supporting phenomenon — if the 2 kg packet has a base area 4 times that of the 1 kg packet, what is the L.S.F? Learners update the DQB: "Area scales as the square of the linear scale factor." Lesson 6 — Volume Scale Factor (V.S.F) and the L.S.F–A.S.F–V.S.F Relationship: Learners handle (or calculate dimensions of) pairs of similar solids — the sufuria cooking pots and the two school water tanks from the anchoring phenomenon. They tabulate L.S.F, A.S.F, and V.S.F and establish V.S.F = $(\text{L.S.F})^3$. The group that predicted the water-tank capacity in Lesson 1 now confirms or corrects their prediction using V.S.F. Learners formalise the three-way relationship and record it on the DQB. Lesson 7 — Application to Real-Life Situations & Model-Building Project Launch: Learners work through real-life problems involving maps (Survey of Kenya scale), architectural plans, and packaging design using L.S.F, A.S.F, and V.S.F. The project is formally launched: groups select a solid (cube, cuboid, cylinder, or sphere), choose a scale factor, and use locally available materials (cardboard, clay, bottle caps, maize cobs) to construct a set of at least two similar models. Groups plan their models, calculate required dimensions, and begin construction. Lesson 8 — Project Presentations, Model Exhibition & Unit Consolidation: Groups complete and present their similar-solid models, explaining the L.S.F used and demonstrating the corresponding A.S.F and V.S.F. The class holds a mini-exhibition. Learners

	revisit the anchoring phenomenon (school plan and water tanks) and articulate full answers to the driving question. The DQB is reviewed collaboratively, remaining questions are addressed, and a cumulative annotated diagram model (object → image with all three scale factors labelled) is drawn individually as a consolidation activity.
Supporting Phenomena	– A packet of Unga Pembe maize flour comes in three sizes — 1 kg, 2 kg, and 5 kg — whose packaging boxes are similar in shape. Learners measure the boxes and find that the 2 kg box is not simply twice as wide as the 1 kg box; they explore how the manufacturers calculated the dimensions so that the volume (and hence the flour capacity) scales correctly. – Google Maps / a printed Survey of Kenya topographic map of the Rift Valley shows Lake Nakuru National Park at a stated scale. Learners measure a distance on the map and calculate the actual ground distance, directly applying linear scale factor in a geographic Kenyan context. – Photographs of a Maasai junior warrior (moran) standing next to a scale model of a traditional manyatta homestead reveal that the model is geometrically similar to the real homestead. Learners identify corresponding lengths and calculate the scale factor used by the craftsman. – Two similar cylindrical metal cooking pots (sufuria) of different sizes are displayed: one is used for cooking ugali for a family of 5, the other for a school kitchen feeding 200 learners. Learners predict how many times more water the larger pot holds given that its diameter is 3 times that of the smaller one, confronting the volume scale factor concept.

LESSON 1 (80 minutes): Anchoring Phenomenon & Properties of Similar Figures	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor learners in the phenomenon of the school architectural plan and the two water tanks, and to establish the properties of similar figures as the foundation for the sub-strand.
Knowledge	Learners determine the properties of similar figures: equal corresponding angles and proportional corresponding sides (Linear Scale Factor).
Skills	Learners collect and sort objects from the school environment into similar and non-similar sets, measure corresponding sides, and calculate ratios to identify proportionality.
Attitudes	Learners appreciate the use of similarity and enlargement in real-life situations, and take responsibility for collecting and caring for objects from the immediate environment to recognise those that are similar.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	This is the opening lesson of the sub-strand. It launches the anchoring phenomenon — the school architectural plan labelled Scale 1:200 and the two water tanks of different sizes on the Kisumu school compound — generating genuine cognitive conflict between learners' intuitive expectation (double the size → double everything) and the mathematical reality that will unfold across the storyline. By the end of this lesson, learners have a shared reference point (the phenomenon), a working definition of similar figures, and an open Driving Question Board (DQB) that will be revisited and built upon in every subsequent lesson.
Safety Notes	Learners handling tins, seed pods, or other collected objects should check for sharp edges or rusty metal and report any to the teacher before use. Wash hands after handling objects from the school compound.

B. LESSON OVERVIEW
Learners are introduced to the anchoring phenomenon: a school architectural plan (Scale 1:200) pinned in the staffroom of a Kisumu school, and two water storage tanks on the compound that are identical in shape but where one holds exactly 8 times as much water despite appearing only twice as tall. The teacher reveals the phenomenon through a printed or projected image of the plan and a photograph of the two tanks. Learners make initial predictions about what happens to lengths, areas, and volumes when a shape is scaled. In groups they then collect objects from the school environment (Kimbo tins, exercise books, maize seed packets, matchboxes) and sort them into similar and non-similar sets. Through guided measurement and ratio-checking, they surface and record the two properties of similar figures: equal corresponding angles and proportional corresponding sides. The lesson closes with the formal posting of the Driving Question on the Driving Question Board (DQB), and learners add their first round of questions generated by the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1 — Predict 📹 **VIDEO:** Geometry and	Learners view the image of the school architectural plan (Scale 1:200) and the photograph of the	1. Display the phenomenon: pin or project a clear labelled image of a school architectural plan marked	Prediction Journaling with Public Commitment: learners commit their intuitive models in writing	Teacher scans individual predict slips during circulation. Key indicator: Can the learner

motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	two water tanks. Working individually in their exercise books, they respond to three prompt questions: (1) The classroom block is 10 m wide but appears as 5 cm on the plan — how do you think the architect made everything fit proportionally on one sheet? (2) If you doubled every length of a small water tank to make a big one, what do you think would happen to its area? And to its volume? Would they also double? Write your prediction and your reasoning. (3) Look at the two objects placed on your desk (a small Kimbo tin and a larger similar tin provided by the teacher). Do you think these are 'the same shape'? What would make two shapes truly 'the same shape' even if they are different sizes? Learners write individual predictions before any discussion. Teacher circulates silently for 3 minutes.	'Scale 1:200' alongside a photograph of two water tanks of visibly different sizes on a Kenyan school compound. Say: 'This is a real architectural plan from a school like ours here in Kisumu. Every classroom block, the water tank, and the football pitch fit on this single A3 sheet. The plan is labelled Scale 1:200. Over there are two water tanks — identical in shape, but one holds exactly 8 times as much water as the other, even though it appears only about twice as tall. Today we begin a journey to understand how that is possible.' 2. Distribute the Predict prompt slip (or write prompts on the board). Say: 'Before we discuss anything with a neighbour, I want your own honest prediction in your book. There are no wrong answers right now — I want to see what you are thinking.' 3. Allow 4 minutes of silent individual writing. Use a visible timer. 4. After 4 minutes, cold-call 3 learners (select one high-confidence, one mid, one who rarely volunteers) to share their prediction for prompt 2: 'If you doubled every length, what happens to area and volume?' Record their exact words verbatim on the board under the heading MY PREDICTIONS — DO NOT ERASE. Say: 'We will come back to these predictions at the end of the lesson and again at the end of the sub-strand.' 5. Wait time: after each learner speaks, wait 12 seconds before responding or calling the next learner, to allow the class to process.	before instruction begins. Recording predictions publicly on the board creates cognitive anchors — when the mathematical reality contradicts the intuition (area scales by k^2, volume by k^3), the contrast is visceral and memorable, driving genuine inquiry rather than passive reception.	articulate a reason for their prediction, even if incorrect? Red flag: learner leaves prompts 2 and 3 blank — teacher stops and asks quietly, 'What is your gut feeling — does area double too, or something else?' Collect one prediction slip per group table to inform pacing of Lesson 2.
2 — Observe 📹 **VIDEO:** Geometry and motion in	In groups of 4–5, learners receive a collection bag containing everyday objects from the school environment: two different-sized	1. Distribute collection bags and Group Observation Charts. Say: 'Your task is to act like a mathematician inspecting these	Hands-On Evidence Gathering with Structured Recording: the Group Observation Chart scaffolds learners to move from	Observe whether groups are computing ratios correctly (dividing larger by smaller consistently) and recording at least two pairs of

[Borromini's San Carlo](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Kimbo tins, two exercise books (A4 and A5), two matchboxes of different sizes, two dried maize cobs of different sizes, and one non-similar pair (a circular bottle cap and a rectangular eraser). Groups also receive a ruler, a protractor, and a Group Observation Chart (A3 paper with columns: Object Pair | Corresponding Angles | Ratio of Side 1 | Ratio of Side 2 | Ratio of Side 3 | Similar? Y/N | Reason). Task 1 — Sort: Groups sort all object pairs into 'similar' and 'not similar' piles based on visual judgement alone, and record their initial sort. Task 2 — Measure and check: For each pair judged 'similar', learners measure at least two pairs of corresponding lengths (e.g., height and diameter of tins, length and width of exercise books) and compute the ratio for each pair of corresponding sides. For objects with measurable angles (exercise books, matchboxes), they measure corresponding angles with a protractor. Task 3 — Record: Groups record all measurements and ratios on the Group Observation Chart and note whether the ratios are equal (or approximately equal) across all pairs of corresponding sides. Task 4 — Return to phenomenon: Groups look again at the architectural plan image. Teacher asks: 'Using what you just measured, why does the entire school compound fit on one A3 sheet but still look perfectly proportional?'	objects. First, sort them by eye — similar or not similar. Then use your ruler and protractor to check whether your eye was right. Record everything on your chart.' 2. Circulate to every group within the first 5 minutes. Pose probing questions: 'You said the two tins are similar — what exactly did you check? Have you measured the ratio of heights AND the ratio of diameters? Are they the same ratio?' 3. At the 12-minute mark, stop the class for a 90-second mid-activity sense-check. Cold-call 2 groups: 'Group 3 — what ratio did you get for the two Kimbo tins?' Record on the board. 'Group 5 — same question.' Say: 'Interesting — both groups got approximately the same ratio for height and diameter. Hold that thought.' 4. Redirect groups that are only checking one measurement: 'A shape is not similar just because one length is in the same ratio. What else must be true? Check all corresponding sides.' 5. At 20 minutes, call attention back. Ask: 'Did any group find a pair that looked similar by eye but the ratios were NOT equal when you measured? Hold up your hand.' Invite one such group to explain. Wait 12 seconds. 6. Post the architectural plan image again. Say: 'Now look at this plan. The scale is 1:200. What does that mean for every single measurement on this plan versus the real building?' Wait 10 seconds. Cold-call 2 learners.	visual/intuitive sorting to quantitative verification. By computing multiple ratios per object pair, learners encounter the defining property of similarity (all corresponding side ratios equal; all corresponding angles equal) through their own data rather than from a definition given by the teacher. Returning to the phenomenon at the end of the observe phase connects the abstract measurement activity back to the concrete Kisumu school context.	corresponding sides. Key indicator of understanding: group spontaneously checks a second ratio to confirm similarity without being prompted. Key misconception to catch: group says 'similar because they look the same shape' without measuring — redirect immediately. Collect one Group Observation Chart per group at end of lesson for teacher review.
3 — Explain 📹 **VIDEO:** [Geometry and](#)	Groups share their findings in a whole-class gallery walk format: each group's A3 Observation	1. Pin all Group Observation Charts on the classroom wall or board. Distribute 2 sticky dots per	Gallery Walk + Socratic Questioning + Formalisation: the gallery walk leverages peer data	During the guided example, circulate and check: (a) learner divides 10÷4 = 2.5 AND 15÷6 = 2.5

motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Chart is pinned to the wall. Learners do a 5-minute silent walk to read two other groups' charts and place a sticky dot next to any ratio or angle finding they find surprising. The class then reconvenes. The teacher facilitates a whole-class discussion to formalise the two properties of similar figures. Learners then work a short guided example: Given that two rectangles — one measuring 4 cm × 6 cm (like a small ID card) and one measuring 10 cm × 15 cm (like an A5 page) — are claimed to be similar, verify this claim by: (a) computing the ratio of corresponding lengths and the ratio of corresponding widths, and (b) checking that corresponding angles are equal. Learners write a formal conclusion statement: 'Two figures are similar if and only if…' in their own words, then compare with the KICD definition the teacher reveals.	learner. Say: 'Walk silently and read two charts that are not your own. Place a dot next to any ratio or measurement that surprises you or that you want to question. You have 5 minutes.' 2. After the gallery walk, say: 'I saw many dots in one place. Let's talk about what surprised us.' Invite 3 learners (cold-call from different groups) to explain what surprised them. Wait 12 seconds after each. 3. Facilitate whole-class discussion using Socratic questioning: 'So for the two Kimbo tins, Group 2 found that height ratio = diameter ratio = 1.5. Group 4 found the same. What does this tell us must be true for any two similar objects?' Wait 15 seconds. Accept answers. Write emerging properties on the board. 4. Ask: 'What about angles? Group 1, what did your protractor tell you about the corner angles of the two exercise books?' (Answer: both 90°.) 'Does that mean equal corresponding angles is part of the definition?' Wait 10 seconds. 5. Reveal the two formal properties on the board: PROPERTY 1 — Corresponding angles are equal. PROPERTY 2 — Corresponding sides are in the same ratio (the Linear Scale Factor, LSF). Say: 'Write these in your notes exactly as they appear on the board. These two properties must BOTH be true for two figures to be similar.' 6. Pose the guided example (two rectangles) on the board. Say: 'In your exercise book, verify whether these two rectangles are similar. Show all ratio calculations. You have 4 minutes.' Cold-call 2 learners to share working. Correct any errors immediately. 7. Connect back to phenomenon: 'Now look at the	as the primary evidence source, preventing the teacher from being the sole authority. Socratic questioning guides learners to construct the formal definition from their own observations. The guided example with a Kenyan context (ID card dimensions vs. A5 page) consolidates the definition with a concrete, familiar calculation before it is applied to the phenomenon.	(both ratios equal — similar confirmed), AND (b) learner states angles are all 90°. Misconception to catch: learner adds instead of divides (10−4=6, 15−6=9 — 'not the same, so not similar') — this is an additive rather than multiplicative comparison error; address immediately by asking, 'Is similarity about the difference in length, or about how many times bigger?' Exit check: ask 3 learners (cold-call) to state the two properties of similar figures in their own words before moving to DQB phase.

		architectural plan again. The scale is 1:200. Is the image of the classroom block on the plan similar to the real classroom block? How do you know? Tell your partner.' Wait 12 seconds. Cold-call 1 pair.		
4 — Driving Question Board (DQB) Creation 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives 2 sticky notes. On STICKY NOTE 1 (yellow), they write one thing they now understand about the phenomenon that they did not understand at the start of the lesson. On STICKY NOTE 2 (pink), they write one question that the phenomenon has raised in their mind that they still cannot answer — particularly related to what happens to area and volume when lengths scale. Learners post their notes on the class Driving Question Board (a large sheet of paper or section of the board divided into columns: WHAT WE THINK WE KNOW | QUESTIONS WE STILL HAVE | WHAT WE HAVE FIGURED OUT). The teacher reads selected questions aloud and clusters them into themes visible to the class. The formal Driving Question is then posted at the top of the DQB in bold: 'When you scale up a shape or solid — like doubling the size of a water tank or enlarging a building plan — what exactly happens to its lengths, its area, and its volume, and how can we use this to solve real problems in our community?'	1. Distribute 2 sticky notes per learner. Say: 'Yellow note: one thing you now understand about the phenomenon that confused you at the start. Pink note: one question that is still burning in your mind — especially about what happens to area or volume.' Allow 3 minutes of silent writing. 2. Direct learners to post notes on the DQB. Circulate and read notes as they go up. 3. Read 4–5 pink questions aloud (choose questions that represent the key mathematical tensions of the sub-strand — e.g., 'If length doubles, does area also double?', 'Why does the tank hold 8 times as much if it is only twice as tall?', 'How does the architect calculate the scale for every single room?'). Say: 'These are brilliant questions. Every one of these questions will be answered by the time we finish this sub-strand.' 4. Cluster similar questions on the DQB with a marker. Label clusters: LENGTHS, AREAS, VOLUMES, SCALE IN REAL LIFE. 5. Post the formal Driving Question card at the top of the DQB. Read it aloud together as a class. Say: 'This is our compass for the next several lessons. Every time we figure something out, we will come back and update this board.' 6. Point back to the PREDICTIONS recorded at the start: 'We wrote these predictions at the beginning. Do not erase them. We will test every one of them with	Driving Question Board (DQB) — Public Tracking of Growing Understanding: the DQB makes thinking visible and creates a shared classroom narrative. By separating 'what we think we know' from 'questions we still have', the DQB honours partial understanding and validates confusion as productive. Clustering questions under LENGTHS, AREAS, VOLUMES telegraphs the conceptual arc of the sub-strand to learners, giving them a sense of where the journey is heading and why each future lesson matters.	Read all pink sticky notes before the next lesson. Key indicator of productive confusion: learner asks a question about area or volume scaling (shows awareness that something beyond simple doubling might be occurring). Red flag: all learner questions are about procedure ('How do I calculate scale factor?') with no conceptual curiosity — plan to re-show the tank photograph at the start of Lesson 2 and re-pose the cognitive conflict. Count the number of learners who could correctly state both properties of similar figures on their yellow note — use as a readiness indicator for Lesson 2.

		mathematics before this sub-strand is done.'		
5 — Model Building 📹 **VIDEO:** [Geometry and motion in Borromini's San Carlo] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In their exercise books, learners draw their Initial Model: a rough sketch showing the two water tanks side by side, labelled with what they currently know or believe about the relationship between their sizes. They annotate the sketch with: (a) the property of similar figures they have established today, (b) their current (possibly incorrect) prediction of how area and volume change when length doubles, and (c) one question mark (?) next to the part of the model they are least sure about. Learners write the date and label it 'MODEL 1 — Lesson 1'. The teacher explains that they will return to revise this model in every lesson as understanding grows.	1. Say: 'For the last 6 minutes, you are going to draw your first model of the phenomenon. Open your exercise book to a fresh page. Draw the two water tanks side by side — they do not need to be perfect drawings. Label what you know: the properties of similar figures we established today, and your current best prediction for what happens to area and volume.' 2. Say: 'Put a question mark next to anything you drew that you are not yet confident about. That question mark is not a weakness — it is a sign that you are thinking like a mathematician.' 3. Allow 6 minutes of silent individual model drawing. Circulate and observe. Do not correct predictions about area and volume at this stage — these misconceptions are the productive engine of the next lessons. 4. Ask 2 learners (cold-call) to hold up their model and describe it in one sentence. Wait 12 seconds after each. Affirm: 'This is your starting point. By the end of this sub-strand, this model will look completely different — and you will be able to prove it with mathematics.' 5. Say: 'Label this Model 1 — Lesson 1 and keep this page. We will not tear it out.' 6. Close the lesson by pointing at the DQB and the phenomenon image: 'The two tanks are still there on our board. We now know what makes them similar in shape. Our next question is: what is the mathematical rule that connects their lengths, their areas, and their volumes? That is where we go next lesson.'	Initial Model Construction with Deliberate Uncertainty Marking: asking learners to annotate question marks normalises incomplete understanding and frames model revision (not model replacement) as the goal of the sub-strand. The Initial Model also creates a concrete baseline against which learners can measure their own conceptual growth — a powerful self-efficacy tool aligned with the CBC core competency of Learning to Learn. Returning to the phenomenon image as the closing act reinforces that all future model revisions are in service of explaining the real Kisumu school context.	Scan model sketches for: (a) presence of both similarity properties (equal angles, proportional sides) as annotations — this confirms Explain phase learning was retained; (b) nature of area/volume prediction — most learners will predict linear doubling (area doubles, volume doubles), confirming productive misconception is in place and Lesson 2 on Area Scale Factor is well-motivated; (c) question mark placement — learners who mark '?' on the volume relationship are showing the highest level of metacognitive awareness. Record names of learners with no annotations or no question marks for targeted follow-up at lesson start.

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) PHENOMENON ANCHORING — Did the image of the architectural plan and the two water tanks generate genuine cognitive conflict, or did learners treat it as a routine introduction? If learners were not puzzled by the 8× volume claim, plan to bring a physical demonstration in Lesson 2 (e.g., two similar cylindrical containers where you pour water from the small to the large to show it takes 8 fills). (2) PREDICTION QUALITY — Review the collected predict slips. How many learners predicted that area and volume also double when length doubles? This number indicates how many learners hold the additive misconception that Lesson 2 must directly address. If more than 80% hold this misconception, begin Lesson 2 with an explicit return to these predictions. (3) DQB QUESTIONS — Read all pink sticky notes tonight. Do the learner questions cluster around lengths, areas, and volumes as intended? If most questions are procedural ('How do I draw a scale?'), the phenomenon did not generate sufficient conceptual tension — plan to re-show the tank photograph with the specific prompt: 'The tank is twice as tall. Predict: does it hold twice as much, four times as much, or eight times as much water? Commit to a number before we continue.' (4) GROUP DYNAMICS — Did groups divide measurement tasks equitably, or did one learner dominate the ruler and protractor while others watched? Note groups where participation was uneven and restructure roles in Lesson 2 (assign: Measurer, Recorder, Calculator, Reporter — rotate each lesson). (5) CORE COMPETENCY CHECK — Were learners communicating and collaborating effectively during the object-sorting task? Did the gallery walk produce genuine peer-to-peer engagement, or did learners walk past charts without reading? If the latter, add a structured 'two stars and a question' protocol to the gallery walk in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a school architectural plan labelled Scale 1:200 and two similar water tanks where one holds 8 times as much water despite appearing only twice as tall. Learners also measured corresponding sides and angles of everyday objects (Kimbo tins, exercise books, matchboxes) collected from the school environment.
What did I learn?	Two figures are similar if and only if (1) all corresponding angles are equal and (2) all corresponding sides are in the same ratio (the Linear Scale Factor). The scale 1:200 means every length on the plan is 1/200th of the real length, and every measurement on the real building is 200 times the measurement on the plan.
How does this explain the phenomenon?	Learners cannot yet fully explain why one tank holds 8 times as much water when it is only twice as tall, or why doubling every length does not simply double the area and volume. These questions are posted on the DQB and will be explained through the Area Scale Factor (Lesson 2) and Volume Scale Factor (Lesson 3) investigations.

LESSON 2 (40 minutes): Centre of Enlargement and Linear Scale Factor — Finding the Fixed Point	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners discover that all ray lines drawn through corresponding vertices of an object and its enlarged image meet at a single fixed point — the centre of enlargement — and use this to calculate the Linear Scale Factor (L.S.F) as image length ÷ object length.
Knowledge	– Define the centre of enlargement as the single fixed point from which all ray lines through corresponding vertices of an object and its image converge. – State the formula: L.S.F = image length ÷ object length. – Explain that the school architectural plan's scale of 1:200 is a L.S.F of 1/200, meaning every real-world length is multiplied by 1/200 to produce the plan measurement.
Skills	– Draw ray lines through corresponding vertices of an object and its enlarged image to locate the centre of enlargement. – Measure corresponding sides of similar figures and calculate the L.S.F accurately using a ruler and the formula L.S.F = image length ÷ object length.
Attitudes	– Appreciate that mathematical precision in measurement — as used by Kenyan architects and engineers designing school buildings and water tanks in Kisumu — depends on a consistent, reliable scale factor. – Show responsibility by taking accurate measurements and recording results honestly, reflecting the value of integrity in mathematical work.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	In Lesson 1, learners established that similar figures have equal corresponding angles and sides in the same ratio. Lesson 2 now answers the question of *where* an enlargement originates: every enlargement has a fixed centre point, and the ratio of any image length to its corresponding object length gives the L.S.F — the same number that appears as the scale on the Kisumu school's architectural plan (1:200, i.e. L.S.F = 1/200). This advances the storyline from 'what makes figures similar' to 'how enlargements are constructed and quantified.'
Safety Notes	No specific hazards. Ensure rulers and pencils are handled carefully. Remind learners to press firmly but safely when drawing lines across paper.

B. LESSON OVERVIEW

This lesson is the second in the evidence-gathering sequence for the anchoring phenomenon: a Grade 10 class in Kisumu observing their school's architectural plan labelled "Scale 1:200." Having established in Lesson 1 that similar figures share equal angles and proportional sides, learners now investigate *where* an enlargement comes from — the centre of enlargement — and how to calculate the Linear Scale Factor (L.S.F). The lesson uses a concrete, hands-on activity in which learners are given a pre-drawn object (a simple quadrilateral representing a classroom block's floor plan) and its enlarged image on plain paper, and must draw ray lines through each pair of corresponding vertices to discover that all lines meet at a single point.

The cognitive anchor throughout is the phenomenon: the Kisumu school plan was produced by an architect who placed a centre of enlargement and applied a L.S.F of 1/200 to shrink the real building onto an A3 sheet. By physically constructing ray lines and measuring ratios, learners experience the same mathematical logic the architect used. The formula L.S.F = image length ÷ object length is derived from learners' own measurements rather than transmitted by the teacher, ensuring genuine

sense-making. Learners then apply the formula to the school plan context: if the classroom block is 10 m wide and appears as 5 cm on the plan, L.S.F = 5 cm ÷ 1000 cm = 1/200 — exactly the label on the plan. The lesson concludes with learners updating the Driving Question Board with a new, more precise claim: "L.S.F tells us the ratio of any image length to its object length, and for our school plan that ratio is 1/200." The cumulative model is revised to show the centre of enlargement as a labelled point with ray lines, connecting the abstract geometry to the real-world architectural phenomenon. This sets up Lesson 3, where learners will construct their own enlargements given a centre and L.S.F, and begin exploring how the L.S.F relates to area and volume changes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a printed or hand-drawn worksheet showing a small quadrilateral (labelled ABCD, representing the floor plan of a classroom block) and a larger, clearly enlarged quadrilateral (labelled A'B'C'D') placed nearby on the same page. Before any instruction, learners write individual predictions in their exercise books answering: 'If I draw a straight line from vertex A through vertex A' and extend it, where do you think it will go? Do the same lines from B→B', C→C', and D→D' end up anywhere special — do they all meet, or do they go in different directions?' Learners also predict the ratio of one pair of corresponding sides by estimating with their eyes before measuring.	Distribute worksheets face-down. Say: 'Do not turn over until I say so.' Once distributed, say: 'Turn over. Look at the two shapes — ABCD and A'B'C'D'. Do NOT draw anything yet. In your book, write your name and today's prediction: if you drew a straight line from A through A' and kept going, and did the same from B through B', from C through C', and from D through D' — what do you think would happen to those four lines? Would they go in completely different directions, or might they do something interesting?' Allow 10 seconds of silent thinking (WAIT TIME). Then say: 'Also estimate — just by looking — the ratio of side A'B' to side AB. Write your estimate as a fraction or a number.' After 2 minutes of individual writing, cold-call 3 learners: 'Achieng, what is your prediction?' — 'Otieno, do you agree or disagree with Achieng, and why?' — 'Mwangi, what ratio did you estimate?' Record the three predictions on the board under the heading PREDICTIONS. Do not confirm or deny any prediction. Say: 'Let us find out if mathematics agrees with you.'	Predict-Before-Observe (Anticipation Writing): Learners commit a written prediction to paper before gathering evidence. This activates prior knowledge about lines and ratios from Lesson 1, creates cognitive dissonance when results differ from expectation, and gives the teacher a visible baseline of understanding to target during the Observe phase.	Teacher scans written predictions during the 2-minute window, looking for: (1) whether learners anticipate convergence ('they might meet at a point') versus divergence ('they go everywhere'); (2) whether estimated ratios cluster around a single number (evidence of proportional reasoning from Lesson 1) or are random. Learners who predict divergence or give wildly inconsistent ratios need targeted support during the Observe phase.

Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Working in pairs, learners use a ruler to carefully draw a straight line through vertex A and vertex A', extending the line in both directions across the page. They repeat this for B→B', C→C', and D→D'. They observe where the four lines intersect and mark the point with a cross, labelling it O (the centre of enlargement). Next, each learner independently measures the lengths of corresponding sides: A'B' and AB, B'C' and BC, C'D' and CD, D'A' and DA — recording all eight measurements in a neat table in their exercise books. They then calculate the ratio image side ÷ object side for each pair and record the four ratios. Finally, learners measure the distances OA and OA', and OB and OB', and calculate those ratios too.	Say: 'Now you will find out. Using your ruler, draw a straight line that passes exactly through A and A' — extend it past both points. Do the same for B and B', C and C', D and D'. Take your time — accuracy matters here, just like an architect who must be precise.' Circulate immediately. After 90 seconds, pause the class: 'Hands up if your lines are all meeting at the same point.' Note who raises hands and who does not. Crouch beside pairs who are not converging and check their ruler alignment — say: 'Make sure your ruler edge is touching both A and A' exactly before you draw.' After pairs locate point O, say: 'Mark it clearly and label it O.' Then: 'Now measure each pair of corresponding sides and record them in a table with two columns: Object Side and Image Side. Calculate image ÷ object for each pair. Also measure OA and OA', OB and OB' — calculate those ratios too.' Allow 8 minutes. With 2 minutes remaining, say: 'What are you noticing about all your ratios?' Allow 10 seconds WAIT TIME. Cold-call 2 pairs: 'Wanjiku and Kipchoge — what are all your ratios?' and 'Amara and Njoroge — same question.' Probe: 'Are they all the same number or different?'	Guided Discovery with Structured Data Collection: Rather than being told about the centre of enlargement, learners construct it physically and observe the convergence themselves. The structured measurement table ensures learners gather enough data points (four side ratios plus two distance ratios) to identify the pattern confidently, embodying the NGSS Science and Engineering Practice of 'Analysing and Interpreting Data.'	Teacher checks: (1) Do all four ray lines converge at a single point O? If not, learner is misaligning the ruler — intervene immediately with a demonstration. (2) Are the four calculated ratios approximately equal? Acceptable range given ruler measurement error: within ±0.05 of each other. (3) Are the OA'/OA and OB'/OB ratios also equal to the side ratios? This confirms understanding that the scale factor applies to all distances from the centre, not just side lengths. Record names of pairs who show all ratios equal — these learners can act as peer explainers in the Explain phase.
Explain Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza	Learners write a definition of 'centre of enlargement' in their own words in their exercise books, then share with their pair to agree on a joint definition. The class agrees on a class definition. Learners then write the formula L.S.F = image length ÷ object length and apply it to the Kisumu school plan phenomenon: if a classroom block is 10 metres wide (= 1000 cm) and measures 5 cm on the plan, they	Begin: 'In your book, write a definition of centre of enlargement using only what you discovered — do not look ahead in the textbook.' Allow 90 seconds. Say: 'Share with your partner. Agree on one definition between you.' 60 seconds. Then call on 3 pairs to share: 'Fatuma and Abubakar — read me your definition.' Write key phrases from student responses on the board, building toward the	Concept Definition Writing followed by Phenomenon Re-anchoring: Learners generate their own definition before receiving the formal one, which research shows strengthens retention. Immediately re-anchoring to the school plan phenomenon (calculating 5 cm ÷ 1000 cm = 1/200) ensures the abstract formula is grounded in a tangible Kisumu context learners have been investigating since	Teacher checks exercise books for: (1) A definition that includes 'fixed point' and 'ray lines through corresponding vertices' — incomplete definitions lack either element. (2) L.S.F calculation: 5 ÷ 1000 = 0.005 = 1/200 (accept both decimal and fraction forms). (3) Problem (a): 90 m = 9000 cm; 9000 × (1/200) = 45 cm. (4) Problem (b): 0.45 cm ÷ (1/200) = 0.45 × 200 = 90 cm = 0.9 m.

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	calculate the L.S.F and identify it as 1/200, matching the label on the plan. Learners then solve two practice problems: (a) a football pitch at Kisumu Boys High School is 90 m long; what length does it appear on the 1:200 plan? (b) A door on the plan measures 0.45 cm wide; how wide is the real door in metres?	class definition: 'The centre of enlargement is the single fixed point through which all ray lines connecting corresponding vertices of an object and its image pass.' Underline 'single fixed point.' Say: 'Now — how does this connect to the school plan in Kisumu? The plan is labelled 1:200. The classroom block is 10 metres wide. It appears as 5 centimetres on the plan. Calculate the L.S.F.' Allow 15 seconds WAIT TIME. Cold-call: 'Chebet — walk us through your calculation.' If correct, affirm and ask: 'Why do we write it as 1/200 and not 200?' Ensure learners articulate that the image (plan) is smaller than the object (building), so L.S.F < 1. Then say: 'Solve problems (a) and (b) individually — 3 minutes.' Circulate. After 3 minutes, cold-call one learner for each answer. Probe: 'How did you know to multiply or divide by 200?'	Lesson 1. The two practice problems apply the formula in both directions (object→image and image→object), building flexibility.	Learners who reverse the formula in problem (b) need a prompt: 'Is the plan bigger or smaller than the real thing? So should your answer be bigger or smaller than 0.45 cm?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners revisit the class Driving Question Board, which was started in Lesson 1. Each learner writes one sticky note (or a contribution to the class chart) in response to the prompt: 'What do we now KNOW about L.S.F that we did not know before this lesson?' They also write one new question that today's lesson raises for them — particularly about what happens to area and volume when a shape is enlarged. Contributions are grouped on the DQB under two columns: WHAT WE NOW KNOW and NEW QUESTIONS WE HAVE.	Direct learners to the DQB (a large chart on the classroom wall, established in Lesson 1). Say: 'In Lesson 1 we added our first understanding — similar figures have equal angles and proportional sides, and the school plan uses a scale of 1:200. Today we go deeper.' Hand each learner a sticky note (or ask them to write on a small piece of paper). Say: 'Write ONE thing you now know about L.S.F or the centre of enlargement that you did not know at the start of today's lesson. Then on the back, write ONE new question today's work is making you wonder about.' Allow 2 minutes. Collect and read 5–6 notes aloud, placing them on the board. Cluster similar 'know' statements together. Highlight	Knowledge-Claim Charting with Wonder Questions: The DQB functions as a public, cumulative record of the class's growing understanding. Separating 'what we know' from 'new questions' models scientific thinking — every answer generates new questions — and keeps the storyline tension alive. Highlighting learner-generated questions about area and volume creates anticipatory curiosity for Lessons 3 and 4, maintaining engagement across the sequence.	Teacher reads all sticky notes to check: (1) 'Know' statements should reference L.S.F = image ÷ object, centre of enlargement, or the 1:200 connection — vague statements like 'we learned about shapes' indicate superficial understanding and need follow-up. (2) 'Wonder' questions should be moving toward area/volume (evidence of forward reasoning) or revealing misconceptions (e.g. 'Does L.S.F change for different pairs of sides?' — address immediately by pointing back to the consistent ratios found in the Observe phase).

		questions about area and volume — say: 'Several of you are asking — if L.S.F = 2, does the area also double? Does the volume also double? Keep that question visible — it is exactly where we are going in Lessons 3 and 4.' Point to the original driving question displayed above the DQB and say: 'We can now answer part of our big question: when you scale up a shape, the lengths change by the L.S.F. But what about area and volume? The board reminds us — we are not done yet.'		
Model Building Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear Equations: Deep Dish Pizza (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative model they began in Lesson 1 — a sketch of the Kisumu school plan phenomenon showing the school building and its scaled-down plan image. They now add to this model: (a) a labelled point O representing the centre of enlargement, (b) ray lines drawn from O through each corner of the real building and its corresponding corner on the plan, (c) the L.S.F written as 1/200 with the equation: plan length = real length × (1/200), and (d) an arrow and annotation showing that for the 10 m wide classroom block, L.S.F = 5 cm ÷ 1000 cm = 1/200. Learners annotate the model with a one-sentence summary: 'Every enlargement has a centre O; the L.S.F = image length ÷ object length; for our school plan, L.S.F = 1/200.'	Say: 'Open the cumulative model you drew in Lesson 1. Today you are adding three new features.' Write on the board: '(1) Label a point O as the centre of enlargement. (2) Draw ray lines from O through corresponding corners. (3) Write the L.S.F equation and calculate it for the 10 m classroom block.' Allow 4 minutes for learners to update their models. Circulate, checking that: ray lines extend through both the object corner and image corner (not stopping at one), O is clearly labelled, and the L.S.F calculation is shown as a ratio (not just the decimal). After 4 minutes, ask two volunteers to hold up their models. Use the visualiser or walk the model around the room. Say: 'Notice how Amina has drawn O outside the shape — can the centre of enlargement be inside the shape? Can it be on a side? We will explore that in Lesson 3. For now, your model shows the architecture of our school plan — a centre, ray lines, and a scale factor of 1/200.' End with: 'In Lesson 3, you will use exactly this model to construct your own enlargements	Cumulative Model Revision (Iterative Modelling): The ongoing model forces learners to integrate new knowledge (centre of enlargement, ray lines, L.S.F formula) with prior knowledge (similar figures, proportional sides from Lesson 1). Each revision makes the model more complete and more tightly connected to the phenomenon. Showing two peer models publicly normalises diversity in representation while focusing attention on the key features that all correct models must contain.	Teacher checks models for three non-negotiable features: (1) Point O is labelled and positioned such that ray lines through it pass through both object and image vertices (not randomly placed). (2) L.S.F is written as a ratio — 1/200 or 0.005 — with the calculation shown. (3) The equation 'plan length = real length × (1/200)' or equivalent is present. Models missing any of these features indicate the learner has not yet connected the geometric construction (ray lines) to the algebraic expression (L.S.F formula) — these learners should be prioritised for support at the start of Lesson 3.

		— and then we will start asking what happens to AREA when L.S.F changes.'		

D. TEACHER REFLECTION
1. When learners drew ray lines through corresponding vertices, did all pairs in the class successfully locate a single centre of enlargement O? If several pairs had diverging lines, what measurement or alignment issue was most common, and how will I address ruler technique at the start of Lesson 3? 2. Were learners able to articulate WHY the L.S.F is less than 1 for the school plan (because the plan is smaller than the real building)? Did any learners confuse L.S.F = 1/200 with L.S.F = 200? How will I reinforce the direction of the ratio? 3. Did the Kisumu school plan phenomenon remain genuinely connected throughout the lesson, or did the hands-on activity feel disconnected from the phenomenon? At which specific moment did learners most strongly make the connection between their ray-line construction and the architect's scale drawing? 4. Looking at the DQB sticky notes, how many learners are already wondering about what happens to area and volume? Are there any misconceptions visible on the board (e.g. 'area also scales by 1/200') that I need to explicitly address — or strategically leave visible — before Lesson 4? 5. In the Model Building phase, did learners place the centre of enlargement O in varied positions (inside, outside, on a side of the shape), and did this provoke productive questions about where O can be? Am I prepared to address this variety productively in Lesson 3's construction activity?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners drew ray lines through corresponding vertices of an object (ABCD) and its enlarged image (A'B'C'D') and observed that all four lines converged at a single point O — the centre of enlargement. They also measured corresponding sides and calculated consistent ratios (image side ÷ object side), including OA'/OA and OB'/OB, all equal to the same L.S.F.
What did I learn?	Every enlargement has a unique fixed point called the centre of enlargement (O), from which ray lines through corresponding vertices all pass. The Linear Scale Factor (L.S.F) = image length ÷ object length, and it applies consistently to every pair of corresponding lengths — including distances from the centre O to corresponding vertices. For the Kisumu school plan, L.S.F = 5 cm ÷ 1000 cm = 1/200, which is exactly the '1:200' label on the architectural plan.
How does this explain the phenomenon?	The school plan labelled '1:200' was produced using a centre of enlargement and a L.S.F of 1/200: the architect multiplied every real-world length by 1/200 to produce the plan measurement. This is why the 10 m wide classroom block appears as exactly 5 cm on the plan, and why every window, door, and corridor is in exactly the right proportion — the L.S.F is constant for every single measurement in the drawing.

LESSON 3 (80 minutes): Constructing Images under Enlargement on a Plane Surface	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct the image of a polygon under enlargement on a plane surface using a given centre and linear scale factor (positive and negative values), and verify accuracy by checking that corresponding side lengths are in the correct ratio.
Knowledge	Learners will know: (i) how to locate image points under enlargement using a centre of enlargement and a given L.S.F on plain paper; (ii) the difference in position and orientation of images produced by positive versus negative L.S.F values; (iii) how the construction process confirms that all image sides are in the same ratio to their object sides.
Skills	Learners will be able to: (i) draw the image of a given polygon under enlargement given a centre and a positive or negative L.S.F using a ruler and geometrical set; (ii) verify construction accuracy by measuring corresponding sides and computing their ratios; (iii) connect the constructed image to the architectural plan phenomenon by interpreting L.S.F as a scale.
Attitudes	Learners will appreciate the value of precision and integrity in mathematical construction, recognising that accurate measurement underpins real-world applications such as architectural drawing and water-tank design in Kenyan communities.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	In Lesson 1 learners identified properties of similar figures and established L.S.F from an object–image pair. In Lesson 2 they located the centre of enlargement and computed L.S.F from given figures. In Lesson 3 learners now reverse the process: given a centre and an L.S.F, they physically construct the image of a polygon on plain paper — mirroring exactly what the architect did when producing the 1 : 200 school plan pinned in the staffroom. Comparing a positive L.S.F construction (image on the same side as the object) with a negative L.S.F construction (image on the opposite side, inverted) deepens the cognitive model and directly advances the driving question about what happens geometrically when you scale a shape.
Safety Notes	Learners use sharp pencil compasses and rulers with edges. Remind learners to keep compass points directed away from themselves and peers when not in use. Ensure sufficient desk space between pairs to avoid accidental contact with geometrical instruments.

B. LESSON OVERVIEW

Learners work in pairs using rulers and a geometrical set to construct the image of a triangle (representing a classroom block on the school plan) under enlargement, first with a positive L.S.F (e.g. $k = 2$) and then with a negative L.S.F (e.g. $k = -2$), both from a given centre of enlargement on plain paper. They verify accuracy by measuring all corresponding sides and confirming the ratio equals the L.S.F. Pairs compare constructions, identify the effect of a negative scale factor on the position and orientation of the image, and reconnect findings to the Kisumu school architectural plan phenomenon. The lesson advances the driving question by making the abstract idea of scaling physically tangible and geometrically precise.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 —	Each learner receives a plain sheet of paper with a pre-drawn	Display the pre-drawn triangle and centre O on the board (drawn	Prediction journaling with pair comparison. Activating prior	Observable indicator: Learner sketches show some awareness

Predict 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [KCSE 2021 BUILDING AND CONSTRUCTION PP1 PP2 ANSWERS] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	triangle ABC (labelled 'Classroom Block on the school plan') and a marked centre of enlargement O. They are asked to sketch — without measuring — where they think the image triangle A'B'C' will appear if the shape is enlarged by L.S.F = 2 from O, and then to sketch a second rough image for L.S.F = −2. Learners annotate their sketches with a brief written prediction: 'The image will be ___ times as big and will appear [same side / opposite side] of O because ___.' Pairs share predictions with each other before any construction begins.	large enough for the whole class to see). Say: 'Look at this triangle. It represents one of the classroom blocks on the 1 : 200 architectural plan we have been investigating. Before you touch your ruler, I want you to sketch — just rough — where you think the image will land when we enlarge with scale factor 2 from centre O. Then sketch again for scale factor negative 2. Write one sentence explaining your reasoning.' Allow 4 minutes of silent individual thinking. After 4 minutes say: 'Share your sketch with your partner. Do your images land in the same place? Discuss for 2 minutes.' Call on 3 pairs (cold-call by desk number): 'Pair 4 — where did you predict the image for k = 2 lands, and why?' [WAIT 12 seconds.] 'Pair 7 — does your prediction agree? What is different?' [WAIT 12 seconds.] 'Pair 11 — what do you expect will happen when k is negative?' [WAIT 15 seconds.] Record 3–4 different predictions on the board under two columns: 'k = +2' and 'k = −2'. Do not affirm or correct yet. Say: 'We are going to construct the images precisely so the mathematics can tell us who is right.'	knowledge of direction and distance from a centre, learners commit to a claim before evidence is gathered. This creates cognitive conflict — particularly around the negative scale factor — which primes them to notice the key mathematical result during construction.	that O is the reference point (image rays pass through O). Listen for the language 'same side' vs 'opposite side' in pair discussion. Flag any learner who places the image at the same size as the object regardless of scale factor — this indicates a misconception about L.S.F that must be addressed in Phase 3.
Phase 2 — Observe (Construction Activity) 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Pairs follow a step-by-step guided construction process on plain paper. Step 1 (k = +2): Draw a ray from O through each vertex A, B, C. Measure OA with a ruler; mark A' such that OA' = 2 × OA on the same ray beyond the object. Repeat for B' and C'. Join A'B'C'. Step 2: Measure all three sides of ABC and all three sides of A'B'C'. Record in a two-column table: 'Object side (cm)' | 'Image side (cm)'. Compute the ratio image	Distribute plain paper, rulers, sharp pencils, and geometrical sets. Say: 'We are going to construct our enlargements step by step — this is exactly the process an architect or an engineer uses before a plan goes to print.' Project or draw the step-by-step instructions on the board. Circulate continuously. At Step 1, pause the class after 3 minutes and demonstrate on the board how to draw a ray from O through A	Hands-on guided construction with measurement verification. The act of physically drawing rays, measuring, and computing ratios makes the abstract L.S.F definition concrete. Recording measurements in a table builds the habit of using data to confirm or refute a mathematical claim — connecting to the architectural plan where every measurement on paper must correspond precisely to a real-world distance.	Observable indicators: (1) All three ratios in the table equal 2.0 (allow ±0.1 cm measurement tolerance). (2) For k = −2, image triangle is on the opposite side of O and is inverted relative to the object. (3) Learner annotations use the words 'inverted' or 'opposite side'. Intervene if: any ratio deviates by more than 0.2 (likely a ray-drawing error — re-demonstrate measuring from O); image for k = −2 is on the same side as k = +2 (learner has

📖 **PDF:** [KCSE 2021 BUILDING AND CONSTRUCTION PP1 PP2 ANSWERS] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	side ÷ object side for each pair. Step 3 (k = −2): On a fresh diagram with the same O and triangle ABC, draw rays from O through each vertex but this time mark A'' such that OA'' = 2 × OA on the opposite side of O (i.e., the ray extends through O in the reverse direction). Repeat for B'' and C''. Join A''B''C''. Observe and annotate: orientation of image (same / inverted), position relative to O, size. Step 4: Pairs record three observations in their notebooks under the heading 'What I notice about positive vs. negative L.S.F.'	and how to measure OA accurately. Say: 'Place your ruler so its zero mark is exactly at O. Read the distance to A. Now double it and mark that point. That point is A-prime.' [WAIT 10 seconds for learners to apply.] At Step 2, when pairs begin measuring, ask targeted questions to 3 pairs (rotate to different sides of the room): 'What is OA for your triangle? What should OA-prime be?' [WAIT 12 seconds.] 'Is your ratio exactly 2? If not, where might the measuring error have crept in?' [WAIT 12 seconds.] Before Step 3, pause the class and say: 'Now the interesting part. For k = negative 2, the distance from O to the image is still double — but which direction do we travel along the ray?' Cold-call 2 learners: 'Amina — which direction?' [WAIT 15 seconds.] 'Otieno — do you agree or disagree, and why?' [WAIT 15 seconds.] Confirm: 'We travel through O and out the other side — the image is on the opposite side of O from the object.' Monitor that all pairs complete Step 3 correctly before moving to Phase 3.		not understood the direction reversal).
Phase 3 — Explain 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [KCSE 2021 BUILDING AND CONSTRUCTIO	Each learner writes an individual explanation in their notebook answering three prompts: (1) 'Describe in your own words the construction steps for enlargement with a given centre and L.S.F.' (2) 'What is different about the image when L.S.F is negative compared to when it is positive?' (3) 'The Kisumu school architectural plan has a scale of 1 : 200. If you were the architect, what L.S.F would you use to draw the plan from the real building, and what L.S.F would you use to calculate real	After construction, say: 'Put your ruler down. Independently, answer the three prompts on the board in your notebook. You have 6 minutes.' After 6 minutes: 'Share with your partner — 2 minutes.' Reconvene and cold-call 4 learners for prompt responses (one prompt per learner pair, rotating across the room): 'Zawadi — describe the construction steps in your own words.' [WAIT 12 seconds.] 'Kipchoge — what changes when the scale factor is negative?' [WAIT 15 seconds.]	Claim–Evidence–Reasoning (CER) written explanation. Learners state a claim (what L.S.F does), cite evidence from their measurement table, and reason using the formal definition. This strategy builds scientific argumentation habits and ensures the construction activity produces durable conceptual understanding rather than just procedural steps.	Observable indicators: (1) Learner correctly states that negative L.S.F produces an inverted image on the opposite side of O. (2) Learner correctly identifies the architect's L.S.F as 1/200 (or equivalently 0.005) and the reverse as 200. (3) CER explanation connects distances from O to the L.S.F definition. Flag any learner who says 'the image gets smaller when k is negative' — this conflates sign with magnitude; address in the consolidation.

N PP1 PP2 ANSWERS Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	dimensions from the plan?' Pairs then share explanations. The class reconvenes for a whole-class discussion anchored by the teacher.	'Achieng — for the school plan at 1 : 200, what L.S.F did the architect use?' [WAIT 15 seconds.] 'Mutua — if you find a room is 3 cm wide on the plan, how wide is it in real life?' [WAIT 12 seconds.] Consolidate on the board: write the formal definition — 'For an enlargement with centre O and L.S.F k: OA' = k × OA. If k > 0, image is on the same side of O as the object. If k < 0, image is on the opposite side of O and is inverted.' Link explicitly to the phenomenon: 'The architect used k = 1/200 to shrink the real building onto paper. We can reverse this: k = 200 scales the drawing back up to reality. When we see the two water tanks that look the same shape but one is twice as tall — what scale factor connects them in length? [WAIT 12 seconds — cold-call 1 learner.] Good. And what does that mean for their areas and volumes? We will investigate that in coming lessons — add that question to your Driving Question Board.'		
Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** KCSE 2021 BUILDING AND CONSTRUCTIO N PP1 PP2	Each learner adds one sticky note (or written entry in their DQB section of their notebook) in response to two prompts: (A) 'What have I now figured out about scaling shapes?' and (B) 'What new question has today's construction raised for me?' The class then does a 3-minute gallery walk to read each other's new questions. Learners place a tick next to any question they share. The teacher harvests the most-ticked questions and pins them to the class DQB under the heading 'Lengths — SOLVED ✓ | Areas — still to investigate | Volumes — still to investigate.'	Say: 'Update your Driving Question Board. Write what you have now figured out — be specific, use numbers and the term L.S.F. Then write one new question your construction raised.' Allow 3 minutes of silent writing. Facilitate a 3-minute standing gallery walk: 'Walk around, read three peers' questions, tick the ones you share.' Harvest ticks. Read the top two questions aloud. Say: 'We now know exactly what happens to lengths when you scale — the image side equals k times the object side, full stop. But our driving question asks about area and volume too. Look at the two water tanks on our	Driving Question Board curation. Making thinking visible on a shared public board allows the class to track the progression of understanding across the 8-lesson sequence. Categorising questions as 'solved' vs 'still to investigate' models scientific practice — recognising what is known and identifying productive next questions.	Observable indicators: DQB entries use the term 'L.S.F' and reference the construction (e.g. 'I now know that every side of the image is exactly k times the matching side of the object'). New questions mention area or volume (indicating readiness for Lessons 4–5). Any learner whose entry still conflates 'enlargement' with 'only making bigger' should be flagged for individual follow-up.

ANSWERS Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings		phenomenon: one is twice as tall. Does the area of its base double? Does its volume double? Add those unresolved questions to our board. We are going to find out in Lessons 4 and 5.' Pin two new question cards to the DQB: 'If L.S.F = k, what is the Area Scale Factor?' and 'If L.S.F = k, what is the Volume Scale Factor?'		
Phase 5 — Model Building 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** KCSE 2021 BUILDING AND CONSTRUCTION PP1 PP2 ANSWERS Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners update their cumulative 'Scaling Model' diagram that they began in Lesson 1. They add a new layer to their diagram: an annotated construction sketch showing centre O, object triangle, image triangle for k = +2, and image triangle for k = −2, with labelled measurements confirming the ratio. Beneath the diagram they write a one-sentence model statement: 'When a polygon is enlarged with centre O and L.S.F k, every image side = k × (object side), and the image is inverted and on the opposite side of O when k is negative.' Pairs check each other's model statements for accuracy before the lesson closes.	Say: 'Turn to the Scaling Model page you started in Lesson 1. We are going to add today's construction as a new layer. Draw a small annotated sketch — it does not need to be a full construction, just clear enough to show centre O, the two triangles for positive and negative k, and two measurements confirming the ratio.' Allow 5 minutes. Circulate and look for: correct labelling of O, A, A'; arrows showing direction of rays; at least one ratio calculation written on the diagram. Say: 'Swap with your partner. Check: (1) Is O labelled? (2) Are rays drawn from O through each vertex? (3) Does the written ratio equal k? (4) Is the negative-k image on the correct side?' After peer-check say: 'Who found something to correct in their partner's model? Hands up.' Address 2–3 common corrections whole-class. Close by returning to the phenomenon: 'Every time an architect in Nairobi, Kisumu, or Mombasa draws a building plan, they are doing exactly what you did today — applying a centre and a scale factor to every point of the building's outline. The precision you practised with your ruler is the same precision that ensures a water tank is the right size and a doorway is wide enough. Our next step is to discover what that same	Cumulative annotated model revision. By physically adding to a growing diagram across lessons, learners build a visual and written record of their expanding understanding. Peer-checking the model statement introduces collaborative accountability and surfaces residual misconceptions before they fossilise.	Observable indicators: (1) Model diagram correctly shows image on opposite side of O for k < 0. (2) Written model statement includes 'image side = k × object side' and references the sign of k. (3) Peer-check catches at least one inaccuracy per pair (indicating active reading rather than passive agreement). Exit indicator: teacher collects 5 random model pages to check before Lesson 4.

		scale factor does to areas.'		

D. TEACHER REFLECTION
After the lesson, consider: (1) Did all pairs successfully complete both constructions (k = +2 and k = −2) with ratios within acceptable tolerance? If not, was the difficulty with ray-drawing, measurement technique, or the concept of direction reversal for negative k? Plan a 5-minute re-demonstration at the start of Lesson 4 if more than one-third of pairs struggled with the negative L.S.F construction. (2) Did learners' DQB entries show genuine curiosity about area and volume scale factors? If the class is not yet asking 'does area also double?', use the water-tank hook more explicitly at the start of Lesson 4. (3) Were CER written explanations strong enough to demonstrate understanding, or were they mostly procedural descriptions without reasoning? If the latter, introduce a sentence frame in Lesson 4: 'I claim ___ because the evidence from my measurements shows ___, which means ___.' (4) Note which learners were flagged during formative assessment checks — arrange brief targeted support before Lesson 4 for those who conflated magnitude with sign of L.S.F or who could not connect the construction to the 1 : 200 scale phenomenon.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners drew rays from the centre of enlargement O through each vertex of a triangle representing a classroom block on the Kisumu school plan, measured distances from O to each vertex, applied the given L.S.F (k = +2 and k = −2) to locate image vertices, and measured all corresponding sides to verify the ratio.
What did I learn?	When a polygon is enlarged with centre O and L.S.F k, every image side equals k times the corresponding object side. A positive k places the image on the same side of O as the object; a negative k places the image on the opposite side of O and inverts its orientation. The architect's 1 : 200 scale is equivalent to an L.S.F of 1/200 applied to every point of the real building.
How does this explain the phenomenon?	Learners wrote Claim–Evidence–Reasoning explanations connecting their measurement tables to the formal definition OA' = k × OA, identified the effect of the sign of k on image position and orientation, and updated their cumulative Scaling Model diagram to include both positive and negative enlargements anchored to the school architectural plan phenomenon.

LESSON 4 (80 minutes): Enlargement on the Cartesian Plane: Positive and Negative Linear Scale Factors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct images of objects under enlargement on the Cartesian plane, using a specified centre (including the origin) and both positive and negative linear scale factors, and connect this to GPS grid coordinates used in Kenyan land surveying and the Kisumu school architectural plan.
Knowledge	Learners will know: (a) how to plot and enlarge an object on the Cartesian plane given a centre of enlargement and a linear scale factor (L.S.F.); (b) that a positive L.S.F. produces an image on the same side of the centre as the object, while a negative L.S.F. produces an image on the opposite side; (c) how the Kisumu school plan (Scale 1:200) functions as a coordinate grid in which every point can be located using ordered pairs.
Skills	Learners will be able to: (a) accurately plot an object and its image on the Cartesian plane under enlargement with a given centre and L.S.F.; (b) construct enlargements using both positive and negative L.S.F. using a ruler and geometrical set; (c) determine the image coordinates from first principles using the formula: Image position = Centre + L.S.F. × (Object position − Centre); (d) verify their constructions by checking that corresponding lengths are in the ratio |L.S.F.| : 1.
Attitudes	Learners will appreciate: (a) the use of similarity and enlargement in real-life situations such as GPS land surveying, architectural plan reading, and urban planning in Kenyan cities; (b) the value of integrity in making accurate geometric constructions and adhering to correct measurements; (c) the value of responsibility in collaborating to construct and verify geometric images.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	In Lessons 1–3 learners established what similar figures are, how to find the centre of enlargement and L.S.F. from plane figures, and how L.S.F. relates to Area Scale Factor and Volume Scale Factor. This lesson moves the work onto the Cartesian plane — the mathematical language of GPS grids and land survey maps. By treating the Kisumu school architectural plan (Scale 1:200) as a coordinate grid, learners discover that enlargement is a precise, coordinate-driven process, and that reversing the sign of the L.S.F. flips the image through the centre — a result that will later underpin rotation and other transformations. This lesson advances the driving question by showing exactly HOW lengths scale on a coordinate system, preparing learners for the area and volume scale factor generalisations in Lessons 5–6.
Safety Notes	Rulers, compasses, and set squares have sharp edges. Learners should handle geometrical instruments with care and store them flat on desks when not in use. No additional safety concerns for this lesson.

B. LESSON OVERVIEW

Learners begin by predicting where the image of a small triangle (representing a classroom block on the Kisumu school plan) will land on a Cartesian grid when enlarged from a given centre with L.S.F. = 2. They then observe a teacher-led demonstration plotting the image step by step. In the Explain phase, pairs construct their own enlargements — first with a positive L.S.F. then with a negative L.S.F. — and record coordinate patterns. The Driving Question Board is updated to capture the new insight that a negative L.S.F. rotates the image 180° through the centre. The lesson closes with learners revising their cumulative model of the school plan, now annotating it with coordinate pairs to mirror how Kenyan land surveyors use GPS grid references.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (15 minutes) 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions on a Cartesian Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a printed A4 Cartesian grid (axes −10 to 10 in both directions) with triangle ABC already plotted: A(1, 1), B(3, 1), C(3, 4). The grid is labelled 'Kisumu School Plan — Coordinate Grid View'. Learners are told: 'This triangle represents the footprint of one classroom block on the school plan. The centre of enlargement is the origin O(0, 0) and the L.S.F. is 2.' Working individually for 4 minutes, learners sketch — WITHOUT any calculation — where they think the image triangle A'B'C' will appear. They annotate their guess with estimated coordinates. Learners then share predictions with a shoulder partner (2 minutes) and record one point of agreement and one point of disagreement on a sticky note placed on the class Prediction Wall.	Distribute the pre-printed grids. Read the scenario aloud: 'The architects of this Kisumu school drew classroom Block C at grid coordinates A(1,1), B(3,1), C(3,4). A new plan will enlarge the block by a scale factor of 2 from the origin. Where will the new block appear on the grid?' Allow 4 minutes of silent individual work. Circulate and note two or three interesting (including incorrect) predictions without correcting them. After 4 minutes say: 'Share your prediction with your partner. Find ONE thing you agree on and ONE thing that is different.' After 2 minutes of partner talk, cold-call 3 pairs to share: 'Pair by the window — what did you agree on?' (WAIT 12 seconds.) 'Pair near the door — what was different between your predictions?' (WAIT 12 seconds.) Record 3–4 representative predicted coordinate sets on the board in a table labelled 'Our Predictions'. Do NOT confirm or deny. Say: 'We will come back to these predictions to see who was closest — and more importantly, WHY.'	Elicit-then-Refine Prediction: by committing to a coordinate guess before instruction, learners activate prior knowledge of the Cartesian plane and reveal misconceptions (e.g., 'the image just shifts two units' or 'only the x-coordinate doubles'). Recording predictions publicly creates cognitive accountability — learners are motivated to resolve the tension between their guess and the mathematical outcome.	Observable indicator: Do learners double BOTH coordinates (correct intuition) or only shift/scale one axis? Teacher scans written predictions during circulation. If fewer than half the class doubles both coordinates, insert a 2-minute bridging question: 'If a point is at (3, 1) and I scale everything by 2 from the origin, what happens to the distance from the origin to that point?' This reveals whether learners understand that L.S.F. from the origin multiplies the position vector.
Phase 2 — Observe (18 minutes) 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions on a Cartesian Plane	The teacher projects (or draws on the board) a large Cartesian grid and works through the enlargement of triangle ABC with centre O(0,0) and L.S.F. = 2 in real time. Learners follow along on their own printed grids, plotting each image point as the teacher does. The teacher narrates each step explicitly. After the positive L.S.F. demonstration (8 minutes), learners watch a second demonstration with L.S.F. = −2 on the SAME object triangle ABC,	Step 1 — Positive L.S.F. demonstration (8 min): Draw axes clearly. Plot A(1,1), B(3,1), C(3,4). Say: 'To find the image of any point under enlargement from centre O with L.S.F. k, we use: Image = O + k × (Point − O). Since our centre is the origin, this simplifies to: Image = k × Point.' Plot A'(2,2) saying: '2 times (1,1) gives (2,2).' Plot B'(6,2) and C'(6,8). Connect to form triangle A'B'C'. Ask: 'Before I join the points — cold-call: Wanjiku, what do you	Worked Example with Parallel Comparison: the two back-to-back demonstrations (positive then negative L.S.F.) allow learners to isolate the effect of the sign of L.S.F. by holding the centre and magnitude constant. This mirrors the scientific practice of 'changing one variable at a time' and builds structural understanding rather than procedural mimicry. The three observation prompts drive learners to articulate the pattern in their own words before formal	Observable indicator: Can learners correctly plot at least two of the three image points during the demonstration? Teacher circulates after each point is plotted and scans grids. Specific check: Is C''(−6,−8) plotted in the correct quadrant (third quadrant)? A common error is plotting it as (6,8) — same quadrant as C. If this error is widespread, pause and ask: 'If L.S.F. is negative, what sign will the coordinates of the image have if the object is in the

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	centre still O(0,0) (5 minutes). Learners plot this second image on the same grid using a different colour pen. For the final 5 minutes, learners compare the two image triangles (L.S.F. = +2 and L.S.F. = −2) and answer three observation prompts written on the board: (1) What is the same about the two images? (2) What is different? (3) Where is the image relative to the centre when L.S.F. is negative?	notice about where A' is relative to O and A?' (WAIT 12 seconds.) Then: 'Ochieng, does A' lie on the ray from O through A?' (WAIT 10 seconds.) Confirm: 'Yes — the image always lies on the ray from the centre through the object point, at k times the distance.' Compare with the Prediction Wall: 'Which prediction was closest? Why?' Step 2 — Negative L.S.F. demonstration (5 min): Erase the image only. Say: 'Now let us use L.S.F. = −2. Same formula: Image = −2 × Point.' Plot A''(−2,−2), B''(−6,−2), C''(−6,−8) in a contrasting colour. Ask: 'Fatuma, what direction is A'' from the origin compared to A?' (WAIT 12 seconds.) Highlight that A'' is on the OPPOSITE side of O from A. Step 3 — Observation prompts (5 min): Write the three prompts on the board. Give 3 minutes of silent written response, then cold-call 3 learners for one prompt each.	vocabulary is introduced.	first quadrant?' (WAIT 15 seconds.)
Phase 3 — Explain (25 minutes) 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions on a Cartesian Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs. Each pair receives a task card with two enlargement problems set in the Kisumu school-plan context. Problem 1 (positive L.S.F.): The water tank enclosure on the school plan has a rectangular footprint with vertices P(1, 2), Q(1, 4), R(3, 4), S(3, 2). Enlarge it with centre C(1, 2) and L.S.F. = 3. Find and plot the image P'Q'R'S'. Verify by measuring that each image side is 3 times the corresponding object side. Problem 2 (negative L.S.F.): The school gate is represented by triangle E(2, 1), F(4, 1), G(3, 3). Enlarge it with centre O(0, 0) and L.S.F. = −2. Find and plot the image E'F'G'. Describe in one sentence what happened to the orientation of the triangle. After	Distribute task cards. Say: 'You have 15 minutes to complete both problems. For Problem 1, the centre is NOT the origin — think carefully about how the formula changes.' Circulate actively. After 5 minutes, check in with struggling pairs: 'For Problem 1, what is the vector from the centre C(1,2) to the point P(1,2)? What does that tell you about P's image?' (WAIT 12 seconds — this is a deliberate hint that P coincides with the centre, so P' = P.) For confident pairs: 'Can you generalise — if a point lies exactly on the centre of enlargement, what is always true about its image?' After 15 minutes, direct pairs to join another pair. Say: 'You have 5 minutes to compare your General Rule	Collaborative Problem-Solving with Generalisation Press: pairs first generate results concretely (plotting coordinates), then are pressed to abstract a General Rule in words. The two-pair compare-and-agree structure (think-pair-share-square) ensures that multiple phrasings are evaluated, building precision in mathematical language — a Core Competency (Communication and Collaboration). The connection to GPS land surveying grounds the abstraction in a Kenyan real-life context (National Land Commission, boundary beacons), fulfilling the KICD requirement to apply similarity and enlargement to real-life situations.	Observable indicators: (1) For Problem 1 — does the pair correctly identify that P' = P (since P is the centre) and that Q'=(1,8), R'=(7,8), S'=(7,2)? Partial credit if the formula is set up correctly but arithmetic errors occur. (2) For Problem 2 — are E''(−4,−2), F''(−8,−2), G''(−6,−6) correctly plotted in the third quadrant? (3) Quality of General Rule — does it include the formula with variables, and explicitly mention 'opposite side' for negative L.S.F.? Teacher collects one task card per pair at the end of the phase to review before next lesson.

	completing both problems (15 minutes), pairs write a 'General Rule' statement in their exercise books: 'When the centre of enlargement is NOT the origin, the formula is...' and 'When L.S.F. is negative, the image is...' Pairs then join another pair (group of 4) and compare their General Rule statements (5 minutes). Each group of 4 agrees on one best-worded General Rule and nominates a spokesperson to share it (5 minutes class share-out).	statements. Find any differences and agree on the most precise wording.' After group discussion, cold-call 3 group spokespersons: 'Group 1 — read your General Rule for negative L.S.F.' (WAIT 10 seconds.) 'Group 2 — did you word it differently?' (WAIT 10 seconds.) 'Group 3 — which wording is most precise, and why?' Consolidate on the board: 'Image = Centre + L.S.F. × (Object − Centre). When L.S.F. < 0, the image is on the opposite side of the centre, at a distance of |L.S.F.| times the object's distance from the centre.' Explicitly connect to GPS: 'Kenyan land surveyors at the National Land Commission use exactly this logic — every boundary beacon has a grid reference, and scaling a plot on a GPS map is mathematically identical to what you just did.'		
Phase 4 — Driving Question Board (DQB) Creation (10 minutes) 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions on a Cartesian Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes: a GREEN note — 'Something I now understand about how enlargement works on the Cartesian plane that I didn't understand before this lesson'; and an ORANGE note — 'A new question this lesson has raised for me about scaling, coordinates, or the school plan.' Learners post their notes on the class DQB under two columns: 'We Now Know' and 'We Still Wonder'. The class reads the new orange questions aloud (teacher selects 4–5 representative ones) and discusses as a group which questions will likely be answered in Lessons 5 and 6 (Area Scale Factor and Volume Scale Factor). One learner volunteer updates the running 'Story of Our Investigation' chart by adding: 'Lesson 4: We	Distribute sticky notes (or instruct learners to use the corners of their exercise book pages if sticky notes are unavailable). Say: 'Take 3 minutes to write your two notes — green for something you now know, orange for a new question.' After 3 minutes, say: 'Post your notes on the DQB.' Spend 4 minutes reading selected orange questions aloud. Ask the class: 'Which of these orange questions do you think connects to the school plan mystery about the water tank holding 8 times as much water?' Cold-call 2 learners (WAIT 12 seconds each). Facilitate agreement that the next lessons will tackle HOW the scale factor for area and volume differ from the linear scale factor. Say: 'Amina, please update our Story of Our Investigation chart.' Finally,	Driving Question Board with Metacognitive Colour Coding: green/orange sticky notes distinguish consolidated knowledge from generative questions, training learners in the scientific practice of distinguishing what is known from what is still under investigation. Publicly connecting the new orange questions to upcoming lessons gives learners a sense of the overall inquiry arc and sustains motivation — a key feature of phenomenon-anchored storyline teaching.	Observable indicator: Do the green notes contain specific, accurate statements about coordinate-based enlargement (e.g., 'I now know that Image = Centre + k(Object − Centre)') rather than vague statements ('I understand enlargement now')? Do the orange notes show genuine conceptual curiosity about area and volume scaling, or are they procedural ('How do I check my answer?')? Teacher photographs the DQB to track the evolution of class understanding across the 8-lesson sequence.

	can enlarge any shape on a coordinate grid using Image = Centre + k(Object − Centre). Negative k flips the image through the centre.'	point to the Prediction Wall from Phase 1: 'Look back at your original predictions. Who wants to tell us what they got right and what they now understand more deeply?' Cold-call 2 learners (WAIT 10 seconds each).		
Phase 5 — Model Building (12 minutes) 📹 **VIDEO:** [Intro to the coordinate plane]() Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Functions on a Cartesian Plane (Flexbook)]() Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Kisumu School Plan Model' — a large annotated sketch of the school compound that has been built up across Lessons 1–3. In this lesson they add a coordinate grid overlay: they mark the corner of the main school gate as the origin O(0,0), then annotate at least three key features of the plan (e.g., the large water tank, the football pitch corner flag, the staffroom block) with estimated grid coordinates, using the 1:200 scale to convert real distances to grid units (1 grid unit = 2 m on the real compound). Using these coordinates, learners then record one enlargement calculation in the margin: 'If the small water tank footprint has corner at grid point (2, 3) and the large tank is an enlargement with L.S.F. = 2 from the origin, where is the corresponding corner of the large tank?' They annotate the answer on the model. In the final 2 minutes, learners write a one-sentence 'Model Update Statement' at the bottom of their model page: 'The school plan is a Cartesian grid where every enlargement obeys Image = Centre + k(Object − Centre), and the sign of k determines which side of the centre the image appears.'	Say: 'Take out your Kisumu School Plan Model. Today we add coordinates.' Display the reference conversion: '1 grid unit = 2 metres in real life (Scale 1:200, so 1 cm on paper = 200 cm = 2 m; we let 1 cm = 1 grid unit).' Give learners 5 minutes to annotate coordinates for at least three features. Circulate and prompt: 'Kariuki, you placed the staffroom at (5, 8) — what real-world distance from the gate does that represent in metres?' (WAIT 12 seconds — expected answer: 10 m in x-direction, 16 m in y-direction.) After 5 minutes, direct learners to the water tank calculation: 'Small tank corner at (2,3), L.S.F. = 2 from origin — where is the large tank's corresponding corner?' Cold-call 1 learner for the answer (WAIT 12 seconds — expected: (4,6)). Say: 'Write that on your model.' Give 3 minutes for the Model Update Statement. Close the lesson by pointing to the driving question: 'We started by asking — when you double the size of a water tank, does everything double? Today you proved that LENGTHS scale by k. In Lessons 5 and 6 we will prove what happens to the AREA and VOLUME. Hold on to that question.'	Cumulative Model Revision with Coordinate Anchoring: adding a coordinate grid overlay to the existing school plan model is a concrete instantiation of the lesson's abstraction. By assigning real grid references to familiar school features, learners bridge the abstract Cartesian algebra to a tangible, Kenya-specific spatial context. The one-sentence Model Update Statement is a low-stakes writing-to-learn strategy that consolidates the lesson's core insight and creates a revision artefact learners can return to.	Observable indicator: Does the coordinate annotation use the correct scale conversion (1 grid unit = 2 m)? Is the water tank image coordinate (4,6) correctly computed? Does the Model Update Statement include the formula and the reference to the sign of k? Teacher uses a 3-point rubric in the final 2 minutes: 3 = formula, sign of k, and scale connection all present; 2 = formula present but sign or scale missing; 1 = vague or missing. This informs differentiation for Lesson 5.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Were learners able to apply the formula Image = Centre + k(Object − Centre) independently, or did they rely on the teacher demonstration for each step? If most learners needed step-by-step guidance in the Explain phase, plan a 10-minute fluency drill at the start of Lesson 5 using three rapid-fire coordinate enlargement questions before introducing Area Scale Factor. (2) Did the shift from positive to negative L.S.F. generate genuine cognitive conflict, or did learners treat it as a simple rule ('just make coordinates negative')? If learners are merely procedural, introduce a non-origin centre with negative L.S.F. as a challenge problem in the Lesson 5 opener. (3) Were the GPS/land surveying connections meaningful to learners, or did they feel superficial? Consider inviting a parent or community member who works as a land surveyor (many are present in Kisumu's lakeside property surveying industry) for a 10-minute virtual or in-person Q&A in Lesson 6. (4) Review the task cards collected at the end of Phase 3 — identify the two most common errors and begin Lesson 5 by displaying two anonymised worked examples (one correct, one with the most common error) and asking learners to spot and explain the mistake.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	On the Cartesian grid labelled as the Kisumu school coordinate plan, learners observed that when triangle ABC (object in the first quadrant) was enlarged with L.S.F. = 2 from the origin, every image point was exactly twice as far from the origin as the corresponding object point, lying on the same ray. When L.S.F. = −2 was used, the image appeared in the third quadrant — on the OPPOSITE side of the origin — at twice the distance.
What did I learn?	Learners learned that: (1) enlargement on the Cartesian plane follows the formula Image = Centre + k × (Object − Centre); (2) when the centre is the origin, this simplifies to Image = k × Object; (3) a positive L.S.F. places the image on the same side of the centre as the object; (4) a negative L.S.F. places the image on the opposite side of the centre, effectively rotating it 180° through the centre, while preserving shape and scaling lengths by |k|; (5) this is the same mathematical operation used by Kenyan land surveyors assigning GPS grid references to plot boundaries.
How does this explain the phenomenon?	The cognitive conflict between prediction and observation — that negative L.S.F. does not merely 'shrink' or 'shift' the image but flips it through the centre — is explained by the vector nature of the enlargement formula. Because L.S.F. scales the VECTOR from the centre to the object point, a negative k reverses the direction of that vector while scaling its magnitude. This is why the image appears on the opposite side at the same scaled distance. This explanation also advances the driving question: lengths scale by k (or |k| for negative), confirming and extending the linear scale factor work from Lessons 1–3, and setting up the question of what happens to area (k^2) and volume (k^3) in Lessons 5 and 6.

LESSON 5 (80 minutes): Area Scale Factor: Discovering How Area Grows When You Enlarge a Shape	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to discover and apply the Area Scale Factor (A.S.F) by drawing and comparing similar plane figures, tabulating L.S.F against ratio of areas, and deriving the relationship A.S.F = $(L.S.F)^2$.
Knowledge	Learners will know that: (i) the Area Scale Factor (A.S.F) is the ratio of the areas of two similar figures; (ii) A.S.F = $(L.S.F)^2$; (iii) doubling linear dimensions quadruples area, not doubles it; (iv) the L.S.F can be recovered from A.S.F by taking the square root.
Skills	Learners will be able to: (i) draw pairs of similar rectangles and triangles with a given L.S.F; (ii) calculate the areas of both figures; (iii) tabulate L.S.F against the ratio of areas; (iv) identify the pattern A.S.F = $(L.S.F)^2$ from their table; (v) work backwards from a known A.S.F to find L.S.F; (vi) apply A.S.F reasoning to the Unga Pembe flour-packet context.
Attitudes	Learners will appreciate that: (i) scaling a real-world object (a classroom floor, a maize farm plot, a packet label) does not scale all measurements equally; (ii) responsibility in taking accurate measurements leads to reliable mathematical discoveries; (iii) integrity requires reporting the actual ratio found, even when it contradicts an initial intuition.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	In Lesson 5 learners move beyond linear scale factor (established in Lessons 1–4) to discover that area does not scale linearly. By drawing, measuring, and tabulating pairs of similar figures they construct the rule A.S.F = $(L.S.F)^2$ from evidence. The Unga Pembe flour-packet supporting phenomenon gives an immediate real-world test of the rule. This discovery is the essential bridge to Volume Scale Factor in Lesson 6 and resolves the first half of the cognitive conflict in the anchoring phenomenon: the school plan's area relationships now make sense.
Safety Notes	No significant safety concerns. Ensure rulers and compasses are handled responsibly. Learners should not share geometrical sets across groups without wiping — standard classroom hygiene applies.

B. LESSON OVERVIEW

Learners begin by predicting whether doubling the side length of a rectangle doubles, quadruples, or does something else to its area — deliberately surfacing the common misconception. In the Observe phase, pairs of learners draw three sets of similar figures (rectangles and triangles) with L.S.F values of 2, 3, and ½, calculate both areas, and enter results into a structured table. In the Explain phase, groups compare tables, identify the pattern A.S.F = $(L.S.F)^2$, and test it against the Unga Pembe flour-packet scenario (a base area 4 times larger → L.S.F = 2). The DQB is updated with the new confirmed statement. In the Model Building phase, learners revise their cumulative similarity diagram to annotate area relationships alongside the linear ones already recorded. Throughout, all work is anchored to the architectural plan and water-tank phenomenon introduced in Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (10	Learners are shown a sketch on the board: a small rectangle	1. Display the two rectangles (plan vs real classroom) and the scale	Prediction with Accountability (Predict-then-Revisit): by	Observable indicator: teacher scans finger-votes to identify the

minutes) 📹 **VIDEO:** [Equivalent ratios: recipe] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	labelled 'Classroom on Plan — 2 cm × 3 cm' and told the real classroom is 4 m × 6 m (L.S.F = 1:200 from the anchoring phenomenon). The teacher asks: 'The real classroom floor has some area. The same classroom on the plan also has an area. If the L.S.F is 1:200, what do you think is the ratio of the real area to the plan area?' Learners write a personal silent prediction (just a number or expression) on a sticky note or the corner of their exercise book. They are then asked to choose one of three options displayed: (A) the area ratio equals 200, (B) the area ratio equals 200^2, or (C) the area ratio equals $\sqrt{200}$. Learners hold up fingers (1, 2, or 3) simultaneously so the teacher can scan the room.	label '1 : 200' — point to it and say: 'We have been working with this plan since Lesson 1. Today we ask a new question — not about lengths, but about AREA.' (30 seconds). 2. Pose the prediction question verbally and in writing on the board. Give WAIT TIME of 15 seconds of silence before asking for responses. 3. Run the finger-vote. Scan and silently note the distribution — do NOT reveal the answer. Say: 'Interesting — we have all three options in the room. By the end of today, the mathematics will tell us who is right.' 4. Cold-call 3 learners (one from each option) to justify their prediction in one sentence each. Use the prompt: 'Tell me why you chose that option.' 5. Record the three positions on the board under the heading 'Our Predictions — Lesson 5' for later accountability.	committing a prediction to paper and to a public record, learners create a personal stake in the outcome. This activates prior knowledge about area and exposes the dominant misconception (area scales linearly) before instruction, making the contradicting evidence in Phase 2 more cognitively impactful.	proportion choosing option A (linear scaling) — this is the target misconception. If more than 60% choose A, the teacher knows the Explain phase must spend extra time on the contrast between linear and squared scaling. Record the tally mentally or on a small clipboard note.
Phase 2 — Observe (25 minutes) 📹 **VIDEO:** [Equivalent ratios: recipe] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In groups of three, learners carry out a structured drawing-and-measuring investigation. Each group receives a printed (or teacher-drawn) table with three rows (L.S.F = 2, L.S.F = 3, L.S.F = ½) and five columns: Shape, L.S.F, Area of Small Figure, Area of Large Figure, Ratio of Areas (Large ÷ Small). Groups draw: (i) a rectangle of 2 cm × 3 cm and its enlarged image with L.S.F = 2 (i.e. 4 cm × 6 cm); (ii) a right-angled triangle with base 3 cm, height 4 cm and its image with L.S.F = 3; (iii) a rectangle of 6 cm × 4 cm and its reduced image with L.S.F = ½. They calculate areas using $A = l \times w$ (rectangle) and $A = ½ \times b \times h$ (triangle), enter all values in the table, and compute the ratio of areas. Groups then write the L.S.F and the ratio of areas side by side	1. Distribute the blank table (or instruct learners to rule it in their books) and say: 'You are going to collect mathematical evidence — just like a scientist running an experiment. Your job is to draw accurately, calculate carefully, and let the numbers speak.' (1 min). 2. Circulate every 3–4 minutes. At each group, check that the enlarged dimensions are correct (common error: learners enlarge only one dimension). Use the prompt: 'You said L.S.F = 2 — show me which number you multiplied each side by.' Give WAIT TIME of 10 seconds after the prompt before intervening. 3. At the 15-minute mark, cold-call one group to read out their completed row for L.S.F = 2 (area small = 6 cm^2, area large = 24 cm^2, ratio = 4). Ask the class: 'Does	Data-Table Tabulation with Pattern Detection: providing a structured table channels learners' attention to the relationship between two variables (L.S.F and ratio of areas). By computing L.S.F^2 in the final step themselves, learners generate the generalisation inductively from their own data rather than receiving it as a teacher assertion. This is the NGSS Science and Engineering Practice of Analysing and Interpreting Data.	Observable indicators: (i) all three rows of the table are completed with correct area values (rectangle row 1: 6 cm^2 and 24 cm^2; triangle row 2: 6 cm^2 and 54 cm^2; rectangle row 3: 24 cm^2 and 6 cm^2); (ii) learners correctly compute ratios 4, 9, and ¼ respectively. If a group has an incorrect ratio, the teacher checks whether the error is in the area formula or in the dimension scaling — these require different corrective feedback.

	and look for a numerical pattern. A guiding question is written on the board: 'What is the relationship between the number in the L.S.F column and the number in the Ratio of Areas column?'	your row for L.S.F = 2 match this group? Hold up a tick or cross.' Correct any errors immediately. 4. At the 20-minute mark, prompt: 'Now look only at your Ratio of Areas column. Write L.S.F^2 next to each ratio. What do you notice?' Give 10 seconds WAIT TIME. 5. Do NOT state the rule yet — leave that for Phase 3.		
Phase 3 — Explain (20 minutes) 📹 **VIDEO:** [Equivalent ratios: recipe]() Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos]() 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)]() Source: CK-12 [🔍 Search ARES for similar readings]()	Groups share their tables. The teacher compiles a class consensus table on the board. Learners then write and discuss in their groups: 'State in one sentence what happens to the area when you multiply all lengths by a scale factor k.' After groups share, the teacher formalises: A.S.F = (L.S.F)^2. Learners copy the rule and immediately apply it to the Unga Pembe flour-packet context: 'A 2 kg Unga Pembe packet has a base area 4 times that of the 1 kg packet. What is the L.S.F between the two packets?' Learners work individually (5 minutes), then compare answers with a partner. The teacher then connects this back to the anchoring phenomenon: 'On the Kisumu school plan at scale 1:200, by what factor is the area of the football pitch on the plan smaller than the real pitch?' (Expected answer: 200^2 = 40 000). Learners verify: real pitch ≈ 100 m × 60 m = 6 000 m^2; plan representation = 50 cm × 30 cm = 1 500 cm^2 = 0.15 m^2. Ratio = 6000 ÷ 0.15 = 40 000 = 200^2.	1. Build the class consensus table on the board column by column, asking: 'Group 2, what ratio did you get for L.S.F = 3?' Give WAIT TIME of 10 seconds. Cold-call 3 different groups across the three rows. (4 min). 2. Say: 'Now I want each group to write one sentence — no numbers, just words — describing the pattern.' WAIT TIME 15 seconds of silent writing, then cold-call 2 groups. Synthesise their language into: 'The ratio of areas equals the square of the linear scale factor.' Write this on the board in a box. (4 min). 3. Write the Unga Pembe problem on the board with a small sketch of the two packets. Say: 'Work alone first — I want YOUR answer before you discuss.' WAIT TIME 5 minutes of individual work, then say 'Compare with your partner.' Cold-call one learner to give the answer (L.S.F = √4 = 2) and justify it. (6 min). 4. Pose the football pitch extension. Say: 'Use A.S.F = (L.S.F)^2 and the scale 1:200. What area ratio do you expect?' WAIT TIME 15 seconds. Cold-call 2 learners. Verify with actual dimensions. Say: 'This is exactly the plan on the staffroom wall in Kisumu — the plan doesn't just shrink lengths, it shrinks areas by the SQUARE of that factor.' (6 min).	Claim-Evidence-Reasoning (CER): learners are required to state a claim ('A.S.F = (L.S.F)^2'), cite evidence (their table rows), and reason through two application problems (flour packet and football pitch). This NGSS practice of Constructing Explanations ensures that the algebraic rule is not isolated but is connected to multiple concrete contexts, deepening transfer.	Observable indicators: (i) learners correctly state L.S.F = 2 for the flour-packet problem (not L.S.F = 4); (ii) learners correctly compute 200^2 = 40 000 for the football pitch; (iii) in the CER sentences, learners cite their table data as evidence rather than simply restating the rule. Common error to watch for: learners computing L.S.F = 4 (confusing A.S.F with L.S.F) — address immediately by returning to the table.

Phase 4 — Driving Question Board (DQB) Update (10 minutes) 📹 **VIDEO:** [Equivalent ratios: recipe] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class DQB (a poster or section of the chalkboard maintained across all 8 lessons) is updated together. Learners revisit the prediction from Phase 1. The teacher asks: 'Who predicted option B — area ratio = 200^2? Were you right? Who predicted option A — linear scaling? What does the evidence say now?' Learners write a new DQB sticky note (or entry in their books under 'What We Now Know'): 'Area scales as the SQUARE of the linear scale factor: A.S.F = $(L.S.F)^2$.' They also write one new question generated by today's lesson under 'What We Still Wonder': suggested prompt — 'If area scales as the square, does volume scale as the cube?' The teacher pins/writes both contributions on the DQB. The class also revisits the water-tank observation from the anchoring phenomenon: 'One tank is twice as tall. Its volume is 8 times as much — 2^3. Its surface area — what factor do you predict?' (This is a bridge to Lesson 6.)	1. Return to the Phase 1 prediction tally on the board. Say: 'Let's hold ourselves accountable to our predictions.' Point to each option and cold-call one learner from each original group to explain what the evidence showed. (3 min). 2. Direct learners to write the 'What We Now Know' statement. Give 2 minutes of silent individual writing, then invite one learner to read theirs aloud. Refine class wording collaboratively. Post on DQB. (3 min). 3. Pose the bridge question about volume ('If area scales as k^2, what do you predict for volume?'). Give WAIT TIME 15 seconds. Take 3 cold-call responses without confirming. Say: 'Hold that prediction — Lesson 6 will test it.' Post the question on the DQB under 'What We Still Wonder.' (4 min).	Driving Question Board as Cumulative Knowledge Record: the DQB makes the progression of understanding visible across lessons. Revisiting the Phase 1 prediction creates a narrative of learning ('we thought X, but evidence showed Y') which is more memorable than simply recording a formula. The bridge question seeds productive anticipation for Lesson 6, maintaining inquiry momentum.	Observable indicator: every learner has a correctly worded 'What We Now Know' statement in their exercise book. The teacher does a 30-second visual scan of three or four books while learners are writing the bridge question. Correct wording: must include the word 'square' or the expression '$(L.S.F)^2$' — vague statements like 'area gets bigger' are insufficient and should be flagged.
Phase 5 — Model Building (15 minutes) 📹 **VIDEO:** [Equivalent ratios: recipe] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Area Models for Decimal Multiplication (Flexbook)]	Learners return to their cumulative 'Similarity and Enlargement' model diagram, begun in Lesson 1 and revised in each subsequent lesson. Today they add a new annotated layer: alongside the linear dimensions already marked on their similar figures, they now label the area of each figure and draw a double-headed arrow between them labelled '$\times (L.S.F)^2$'. For the school-plan rectangle representing the administration block (introduced in Lesson 1 as 5 cm on the plan, 10 m in reality, L.S.F = 1:200), learners calculate and annotate the plan area and	1. Say: 'Take out your cumulative model from Lesson 1. Today you are upgrading it — not replacing it. Lengths are already there; now add area.' Display a sample annotated diagram on the board (teacher-prepared, showing the double-headed arrow and '$\times (L.S.F)^2$' label). (2 min). 2. Circulate while learners work. At the 8-minute mark, prompt any group that has only added numbers without the algebraic label: 'I can see you wrote 24 cm^2 — but can you write a general label showing HOW you got from one area to the other?' WAIT TIME	Cumulative Model Revision (Progressive Modelling): iteratively annotating the same diagram across lessons makes the conceptual structure of similarity visible as a growing, connected system rather than isolated facts. Writing the 'Model Revision Note' engages metacognition — learners articulate not just what they know but how their knowledge changed, which deepens retention and prepares them for the final model presentation in Lesson 8.	Observable indicators: (i) the cumulative model diagram correctly shows area labels on both figures AND the algebraic annotation '$\times (L.S.F)^2$'; (ii) the 'Model Revision Note' explicitly contrasts the new understanding with the prior linear-scaling belief ('I used to think area doubled when L.S.F = 2, but now I know it quadruples'). The teacher collects two or three books at the end of the lesson for a quick written-work check before Lesson 6.

Source: CK-12 🔍 Search ARES for similar readings	real area, confirming the factor of 40 000. They also add a mini worked example at the bottom of their model page: the Unga Pembe flour-packet sketch with L.S.F = 2 labelled on lengths and × 4 labelled on areas. Finally, learners write a two-sentence 'Model Revision Note' explaining what changed in their understanding today compared to Lesson 4.	10 seconds before assisting. (8 min). 3. At the 12-minute mark, ask learners to write their 'Model Revision Note'. Cold-call 2 learners to read theirs aloud. Listen for: (a) explicit mention of 'square', (b) a contrast with their earlier belief. Affirm correct responses and gently correct imprecise language. (3 min). 4. Close with: 'Your model now captures both length AND area relationships for similar figures. In Lesson 6, you will add the third layer — volume. Keep this model safe.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) What proportion of learners initially chose option A (linear scaling) in the prediction phase — was the misconception as prevalent as expected, and did the tabulation activity sufficiently challenge it? (2) Were learners able to work backwards from A.S.F to L.S.F (the flour-packet problem) without prompting, or did the square-root step cause confusion? If confusion was widespread, plan a short retrieval starter for Lesson 6 where learners practise $\sqrt{(A.S.F)} = L.S.F$ with two or three quick examples before introducing volume. (3) Did the connection to the anchoring phenomenon (the football pitch area ratio of 40 000) land with learners, or did it feel like an add-on? If learners seemed disconnected from the phenomenon, consider opening Lesson 6 with a brief re-viewing of the school-plan photograph and asking learners to narrate the area story themselves. (4) Were the cumulative model diagrams accurate and detailed enough to support learning in Lesson 6? If many diagrams were incomplete, build a 5-minute model-check into the opening of Lesson 6. (5) Note the names of any learners who consistently chose incorrect area values due to formula errors (confusing perimeter with area) — these learners may need a targeted support session on area formulae before the Volume Scale Factor lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners drew pairs of similar rectangles and triangles with L.S.F values of 2, 3, and ½, calculated the area of each figure, and recorded the ratio of the larger area to the smaller area in a structured table.
What did I learn?	The ratio of areas of two similar figures equals the square of the linear scale factor: $A.S.F = (L.S.F)^2$. Doubling all lengths quadruples the area; tripling all lengths multiplies the area by 9; halving all lengths reduces the area to one-quarter.
How does this explain the phenomenon?	When the L.S.F between two similar figures is k, every length scales by k, but area — which is calculated by multiplying two lengths together — scales by $k \times k = k^2$. This is why the base of the 2 kg Unga Pembe flour packet has an area 4 times (not 2 times) that of the 1 kg packet, even though each linear dimension is only twice as large. The same principle means the Kisumu school plan at scale 1:200 represents areas at a factor of $200^2 = 40\ 000$ smaller than reality.

LESSON 6 (80 minutes): Volume Scale Factor and the LSF–ASF–VSF Relationship	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners establish the Volume Scale Factor (VSF) of similar solids and formalise the three-way relationship VSF = $(LSF)^3$, ASF = $(LSF)^2$, thereby resolving the anchoring phenomenon's water-tank puzzle and equipping learners to solve real-life scaling problems involving capacity and volume.
Knowledge	Learners will know that: (i) VSF = $(LSF)^3$ for any pair of similar solids; (ii) the three scale factors are related by LSF : ASF : VSF = $k : k^2 : k^3$; (iii) doubling a linear dimension multiplies volume by 8, not 2.
Skills	Learners will be able to: (i) calculate VSF from given or measured dimensions of similar solids; (ii) tabulate LSF, ASF, and VSF for multiple pairs of similar solids; (iii) apply VSF to find unknown capacities or volumes of similar containers in real-life contexts.
Attitudes	Learners will appreciate the use of similarity and enlargement in real-life situations (KICD verbatim), particularly in engineering, architecture, and water management in their community.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	In Lesson 6 learners handle (or compute dimensions of) pairs of similar solids — sufuria cooking pots and the two school water tanks from the anchoring phenomenon. They tabulate LSF, ASF, and VSF, establish VSF = $(LSF)^3$, and the group that predicted the water-tank capacity in Lesson 1 now confirms or corrects their Lesson 1 prediction. The three-way relationship is formalised and added to the Driving Question Board.
Safety Notes	No hazardous materials are used. If physical water is poured during the Observe phase, ensure floors are kept dry to prevent slipping. Assign a 'safety monitor' in each group to manage spills. Handle sufuria lids and measuring cylinders carefully to avoid breakage.

B. LESSON OVERVIEW

Lesson 6 of 8 in the Similarity and Enlargement sequence. Learners revisit the anchoring phenomenon — the two water tanks at their Kisumu school that look identical in shape but where one holds exactly 8 times as much as the other despite being only twice as tall. Building on their established understanding of LSF (Lesson 4) and ASF (Lesson 5), learners now investigate Volume Scale Factor (VSF) by working with pairs of similar solids (sufuria cooking pots of two sizes, and the school water tanks represented by scale models or calculated data). They tabulate LSF, ASF, and VSF across multiple pairs, notice the cubic pattern, and formalise VSF = $(LSF)^3$. The group that made a Lesson 1 prediction about tank capacity now revisits and confirms or corrects that prediction. The lesson closes with learners updating their cumulative models and the Driving Question Board with the complete LSF–ASF–VSF relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict 📹 **VIDEO:** Function notation	Learners are shown a projected image of two sufuria (cooking pots) side by side: a small one used at home and a large catering sufuria	1. Display the sufuria image and read aloud: 'The large sufuria has every dimension exactly 3 times the small one — it is a similar solid	Think–Pair–Share with Prediction Recording. Externalising prior knowledge before instruction creates cognitive conflict (learners	Teacher scans written predictions during the 3-minute silent period — if more than 60% predict ×3 (the most common misconception), the

example Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	used at school during lunch. The teacher reveals that the large sufuria has a diameter exactly 3 times the small one, and that both are geometrically similar. In their exercise books learners individually write predictions answering: 'The large sufuria holds __ times as much ugali as the small one. I think this because…' Learners who predicted water-tank capacity in Lesson 1 are reminded to note their original prediction for comparison later. After 3 minutes of silent writing, learners share predictions with a shoulder partner (Think–Pair) before three predictions are cold-called.	with LSF = 3.' 2. Pose the prediction prompt: 'Before we do any maths, write your prediction: how many times more ugali can the large sufuria hold? Write a number AND a reason.' 3. Allow 3 minutes silent thinking — do NOT accept shouted answers. 4. Say: 'Now tell your partner your prediction and one piece of evidence from our previous lessons.' Allow 90 seconds. 5. Cold-call exactly 3 pairs (select one who likely predicts ×3, one who predicts ×9, one who predicts ×27) and record all three predictions on the board under the heading 'Our Predictions — we will test these today.' 6. Ask: 'Which of these connects back to our water-tank puzzle from Lesson 1?' Wait 10 seconds. Cold-call one learner.	holding ×3, ×9, or ×27 sit with uncertainty) and gives the teacher immediate diagnostic information about which misconception dominates the class.	teacher notes this and deliberately sequences the cold-call to surface that prediction first, saving ×27 for last to maximise cognitive conflict.
Phase 2 — Observe 📹 **VIDEO:** Function notation example Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 receive a 'Similar Solids Data Card' with dimension data for three pairs of similar solids drawn from Kenyan contexts: (A) two cylindrical sufuria — small: diameter 20 cm, height 15 cm; large: diameter 40 cm, height 30 cm; (B) two cuboid water tanks — Tank 1: 1 m × 1 m × 1 m; Tank 2: 2 m × 2 m × 2 m (matching the school tanks in the phenomenon); (C) two similar conical grain storage structures (similar to those seen in Siaya County) — small: radius 0.5 m, height 1.2 m; large: radius 1.5 m, height 3.6 m. For each pair learners: (i) calculate LSF; (ii) calculate ASF using the formula from Lesson 5; (iii) calculate the volume of each solid using the correct volume formula; (iv) calculate VSF = Volume(large) ÷ Volume(small); (v) record all four values in a shared tabulation table drawn in their exercise books.	1. Distribute Data Cards and say: 'Your task has five columns — LSF, ASF, Volume(small), Volume(large), VSF. Use the volume formulas you know: cylinder = $\pi r^2 h$, cuboid = l×w×h, cone = $\frac{1}{3}\pi r^2 h$. You have 18 minutes.' 2. Circulate — after 5 minutes, pause ALL groups: 'Everyone stop — check: for Pair A, what did you get for LSF? Show me on your whiteboards / exercise books.' Wait 10 seconds. Cold-call one group. If LSF = 2 is confirmed, say 'Good — now keep going and see what VSF gives you.' 3. For any group stuck on cone volume, ask a guiding question: 'What fraction of a cylinder is a cone with the same base and height?' Wait 10 seconds. Do NOT give the answer. 4. After 15 minutes, ask: 'Without telling me the number yet — is VSF bigger than ASF, smaller, or the same? Write that	Structured Data Tabulation with Cross-Pair Pattern Noticing. By computing three different pairs of similar solids (different shapes, different LSF values: 2, 2, 3) and placing LSF, ASF, VSF side by side in one table, learners are positioned to notice the multiplicative pattern (VSF = LSF^3) through their own data rather than being told. This is aligned with NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data).	Teacher checks each group's tabulation table during circulation. Observable indicator: VSF column should read 8, 8, and 27 for pairs A, B, C respectively. Any group with VSF = 3, 3, 9 (confusing VSF with LSF) is flagged and the teacher asks: 'You found LSF = 3 for Pair C. If you cube 3, what do you get?' — this is a targeted re-direct without whole-class interruption.

	Groups that finish early calculate the actual capacity of Tank 2 in litres and compare with the Lesson 1 prediction posted on the DQB.	observation in your book.' 5. At 18 minutes, signal stop. Invite one group to present their completed table using the board. 6. Reveal the connection to the phenomenon: 'Look at Pair B — that IS our school water tank. What does your VSF tell us about the 8-times capacity claim from Lesson 1?'		
Phase 3 — Explain 📹 **VIDEO:** Function notation example Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups examine their completed tables and answer three guided sensemaking questions written on the board: (Q1) 'For each pair, what is the relationship between LSF and VSF? Try expressing VSF in terms of LSF using an index.' (Q2) 'We already know ASF = $(LSF)^2$. How does VSF compare to ASF?' (Q3) 'Write the three-way relationship in one line using k for LSF.' After 5 minutes of group discussion, whole-class share-out produces the formalised rule: if LSF = k, then ASF = k^2 and VSF = k^3. Learners copy this into their books as the 'Scale Factor Law.' They then apply it immediately: given that a small cylindrical water harvesting tank in Kibera holds 500 litres and a geometrically similar large tank has LSF = 4, calculate the capacity of the large tank. Learners solve individually (3 minutes), then verify with a partner.	1. Write Q1, Q2, Q3 on the board and say: 'Use your table to answer these three questions as a group. You have 5 minutes — I want a written answer to each.' 2. After 5 minutes, cold-call one group for Q1: 'Group at the back — what did you find?' Wait 10 seconds before accepting the answer. Probe: 'Can you write it using an index number?' 3. For Q2, cold-call a different group: 'So if ASF = k^2 and VSF = k^3, what is the relationship between them?' Accept 'VSF = k × ASF' or equivalent. 4. Cold-call a third group for Q3. Write the Scale Factor Law on the board: 'LSF = k → ASF = k^2 → VSF = k^3.' Say: 'This is the three-way relationship KICD wants you to know. Copy it into your Scale Factor Law box.' 5. Pose the Kibera tank problem. Say: 'Work alone. You have 3 minutes.' After 3 minutes: 'Compare with your partner. If you disagree, convince each other.' 6. Cold-call one learner to explain their working aloud step by step. Expected answer: VSF = 4^3 = 64; large tank = 500 × 64 = 32 000 litres. 7. Connect explicitly to phenomenon: 'Now look at Lesson 1's tank puzzle — LSF = 2, so VSF = 2^3 = 8. The tank holds 8 times as much. Who had that prediction correct today?' Acknowledge the correct predictors from Phase 1.	Guided Generalisation from Data (Pattern → Rule → Application). Learners move from their own tabulated data through three structured questions to arrive at the algebraic generalisation themselves. The immediate application to the Kibera tank problem consolidates the rule in a real Kenyan community context, supporting transfer. This reflects NGSS Practice 6 (Constructing Explanations).	Kibera tank calculation provides a whole-class written formative check. Observable success criterion: learner correctly identifies VSF = 4^3 = 64 and multiplies 500 × 64 = 32 000 litres. Common error to watch for: learner uses VSF = 4^2 = 16 (confuses VSF with ASF) — teacher addresses this by asking: 'Which scale factor applies to three-dimensional volume — the square or the cube of LSF?'

Phase 4 — Driving Question Board (DQB) Update 📹 **VIDEO:** [Function notation example] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparing Standard Units of Volume (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front. Two representatives from the group that made the original Lesson 1 water-tank capacity prediction come to the board. They retrieve their Lesson 1 prediction sticky note (or written record), read it aloud, and now use the Scale Factor Law to confirm or correct it: 'We predicted the large tank holds __ times as much. We now know VSF = $(LSF)^3 = 2^3 = 8$, so the large tank holds 8 times as much.' They update the DQB card. The whole class then adds one new sticky note to the DQB responding to the prompt: 'What does today's lesson add to our answer to the driving question?' Each group's note must include the phrase 'VSF = $(LSF)^3$' and one real-life example. Selected notes are read aloud and pinned to the 'What we now know' column of the DQB. Finally, learners identify any remaining questions for the 'What we still wonder' column.	1. Invite the Lesson 1 prediction group to the front: 'Your prediction from Lesson 1 is being tested today. Read us your original prediction.' Wait while they retrieve and read it. 2. Ask them: 'Using what you discovered today, is your prediction correct?' Wait 10 seconds. Allow the group to self-correct if needed — do NOT correct for them. 3. Say to the class: 'Every group write a new DQB sticky note. It must have: (1) VSF = $(LSF)^3$, (2) one real example from Kenya — a tank, a pot, a grain store, a building.' Allow 3 minutes. 4. Cold-call 3 groups to read their notes before pinning. Ask after each: 'Does this bring us closer to answering our driving question about what happens when you scale up a tank?' 5. Prompt the 'still wondering' column: 'What about shapes that are NOT similar — can we still use VSF? Write that as a question on a different coloured note.' 6. Update the DQB 'progress thermometer' (a visual scale 1–8 showing which lesson the class is on) to show Lesson 6 complete.	Driving Question Board Curation with Prediction Revisit. Returning to a Lesson 1 prediction publicly and resolving it with new mathematics gives learners a concrete experience of scientific knowledge building. The sticky-note protocol ensures every learner articulates the new understanding in their own words before it is pinned — preventing passive reception. This mirrors NGSS Practice 7 (Engaging in Argument from Evidence).	Teacher reads DQB sticky notes as they are written. Observable indicator: every note correctly states VSF = $(LSF)^3$ and includes a plausible Kenyan real-life example. Notes that state VSF = $(LSF)^2$ are intercepted before pinning — teacher asks the group: 'We said ASF uses the square — what does VSF use?' and allows self-correction.
Phase 5 — Model Building 📹 **VIDEO:** [Function notation example] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparing Standard Units of Volume (PLIX)] Source: CK-12	Learners retrieve their cumulative 'Scale Factor Model' diagram started in Lesson 4, which currently shows LSF → ASF with the relationship ASF = $(LSF)^2$. They now extend the model by: (i) adding a third layer labelled 'VOLUME (3D)' with the relationship VSF = $(LSF)^3$; (ii) annotating each layer with a Kenyan example (Lesson 4: architectural plan scale, Lesson 5: painted surface area of a tank, Lesson 6: water storage capacity); (iii) drawing arrows from LSF to ASF to VSF labelled '× k' and '× k'	1. Say: 'Take out your Scale Factor Model from Lesson 4. Today you add the final layer — Volume.' Display a projected template showing the three-layer structure. 2. Say: 'You have 7 minutes. Your model must show: three layers (Length, Area, Volume), the formula for each scale factor, one Kenyan real-life example per layer, and arrows showing how they connect.' 3. Circulate — check that the multiplicative arrows (× k between layers) are present. If missing, ask: 'If ASF = k^2 and you multiply by k	Cumulative Model Revision with Peer Feedback. The model is a living artefact that grows across all 8 lessons, making the knowledge-building process visible to learners. Adding the VSF layer today completes the three-way relationship and directly answers the volume half of the driving question. Peer feedback using a structured prompt (one strength, one improvement) develops metacognitive awareness. This reflects NGSS Practice 2 (Developing and Using Models).	Teacher collects 4–5 models at random to review before Lesson 7. Observable success criterion: all three layers present with correct formulas (k, k^2, k^3), arrows connecting them, and at least one Kenyan example per layer. The exit task (Eldoret silo: VSF = 5^3 = 125; volume = 12 × 125 = 1 500 litres) is checked at the start of Lesson 7 as a warm-up and provides a whole-class data point on retention.

🔍 Search ARES for similar readings	to show the multiplicative chain; (iv) writing the complete Scale Factor Law as a boxed statement at the bottom: 'For similar figures/solids with LSF = k: LSF = k, ASF = k^2, VSF = k^3.' Learners share their updated model with their group for peer feedback using the prompt: 'Can you see all three scale factors clearly? Is the Kenyan example specific enough?'	again, what do you get?' Wait 10 seconds. 4. At 5 minutes, prompt peer feedback: 'Show your model to your group. Peer reviewer: find ONE thing that is clear and ONE thing that could be improved.' Allow 2 minutes of peer feedback. 5. Close the lesson: 'By the end of Lesson 8 this model will answer our full driving question. Today you added the volume piece — the piece that explains why our school's large water tank holds exactly 8 times as much as the small one, even though it is only twice as tall.' 6. Set exit task (to be checked at start of Lesson 7): 'A cylindrical grain silo in Eldoret has LSF = 5 compared to a model silo. The model holds 12 litres. Calculate the volume of the real silo in litres.'		

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the majority of learners arrive at VSF = $(LSF)^3$ through their own tabulation data, or did they need to be told? If the latter, plan a short re-engagement at the start of Lesson 7 using a different pair of similar solids (e.g., two similar spherical mandazi dough balls). (2) Which misconception was most persistent — confusing VSF with LSF (×k) or with ASF (×k^2)? Adjust the Lesson 7 warm-up accordingly. (3) Was the Lesson 1 prediction-revisit moment genuinely surprising or flat? If learners had already resolved the tank puzzle informally, consider raising the cognitive demand in Lesson 7 by introducing a frustum or composite solid. (4) Were all groups able to compute cone volumes confidently? If not, a 5-minute cone-volume micro-review may be needed before Lesson 7's surface-area-and-volume synthesis work. (5) Check DQB sticky notes for quality of Kenyan examples — shallow examples (e.g., 'a box') should be upgraded to community-relevant ones (water tanks, grain stores, biogas digesters) in a whole-class share at the start of Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners calculated volumes of similar solids (sufuria cooking pots, school water tanks, and conical grain stores) and tabulated LSF, ASF, and VSF for three pairs of similar solids with LSF values of 2, 2, and 3.
What did I learn?	VSF = $(LSF)^3$ for any pair of similar solids; the complete three-way relationship is LSF = k, ASF = k^2, VSF = k^3; doubling linear dimensions multiplies volume by 8, not 2 — confirming the school water-tank puzzle from the anchoring phenomenon.
How does this explain the phenomenon?	The large school water tank holds exactly 8 times as much as the small tank because LSF = 2 and VSF = 2^3 = 8. The Scale Factor Law (k, k^2, k^3) governs all similar solids and can be used to calculate unknown capacities in real-life contexts such as water storage tanks, cooking pots, and grain silos across Kenya.

LESSON 7 (80 minutes): Application to Real-Life Situations & Model-Building Project Launch	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners apply L.S.F, A.S.F, and V.S.F to solve real-life problems involving Survey of Kenya maps, architectural plans, and packaging design, then formally launch a group model-building project using locally available materials.
Knowledge	Learners know that L.S.F = k, A.S.F = k^2, and V.S.F = k^3; can convert between map/plan distances and real distances; and understand that similar solids preserve shape while scaling lengths, areas, and volumes by the respective scale factors.
Skills	Learners can: (i) use a given map or plan scale to calculate real and plan dimensions; (ii) apply A.S.F and V.S.F to solve area and volume problems in context; (iii) plan and calculate dimensions for a set of at least two similar solid models; (iv) begin constructing models using cardboard, clay, bottle caps, or maize cobs.
Attitudes	Learners appreciate the use of similarity and enlargement in professional fields such as surveying, architecture, and packaging design, and take responsibility for accurate measurements in their own model-building projects.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	This lesson is the payoff lesson: learners move from understanding the mathematics of scale factors (Lessons 1–6) to using those relationships to solve genuine real-world problems — maps, building plans, and water-tank design — exactly like the architectural plan and the two water tanks that sparked their curiosity at the start of the unit. The model-building project formally launches here, giving each group a concrete, hands-on challenge that will produce physical evidence for the cumulative unit model.
Safety Notes	When cutting cardboard or using scissors during the project launch construction phase, learners must point scissors downward when passing them, keep fingers clear of cutting edges, and work on a flat stable surface. Clay and other materials should be handled hygienically. No sharp craft knives are to be used without direct teacher supervision.

B. LESSON OVERVIEW

Learners begin by predicting how the Scale 1 : 200 plan from the anchoring phenomenon compresses and preserves the school compound, then work through three structured real-life problem sets — (1) a Survey of Kenya map, (2) the school architectural plan, and (3) a packaging/water-tank design challenge — applying L.S.F, A.S.F, and V.S.F in each context. Through a jigsaw activity, groups share solutions and consolidate the relationships. In the final phase the model-building project is formally launched: groups select a solid, agree on a scale factor, calculate all required dimensions, record a construction plan, and begin building using locally available materials. The lesson closes with DQB updates and model diagram sketches.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (10 minutes)	Each learner silently examines a printed copy (or teacher-drawn board diagram) of a simplified school plan labelled 'Scale 1 : 200'	Display the plan (or board diagram) and say: 'Before we calculate anything together, I want every one of you to commit to an	Commitment-before-instruction prediction: By requiring a written numerical commitment before any teaching, the teacher activates	Observable indicator: Every learner has three written numerical predictions in their book. Teacher scans for the proportion predicting

📹 **VIDEO:** [Geometry and motion in Borromini's San Carlo] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	— the same plan from the anchoring phenomenon. They individually write predictions in their exercise books answering three prompts: (a) The plan shows a classroom block as 4 cm long. What is the real length of the block? (b) If the football pitch measures 25 cm × 15 cm on the plan, what is its real area? (c) One water tank on the plan appears identical in shape to another but has a base diameter twice as large. How many times more water does the bigger tank hold? Learners commit to numerical answers before any discussion.	answer. No erasing — write your best prediction right now.' Allow 4 minutes of silent individual work. Circulate and note 3–4 varied responses for (c) — expect answers of ×2, ×4, and ×8. After 4 minutes say: 'Turn to your neighbour and compare ONLY your answer to part (c). Do not change your book yet.' Allow 90 seconds. Cold-call 3 learners: 'Akinyi, what did you predict for the tank volume?' — 'Otieno, did you agree or disagree with Akinyi, and why?' — 'Njeri, whose reasoning feels more convincing right now?' Wait 10 seconds after each response before responding or redirecting. Record the three predicted answers to (c) on the board under the heading 'Our Predictions — We Will Test These Today.'	prior knowledge of scale factors (from Lessons 1–6) and surfaces misconceptions (doubling length → doubling volume) that the lesson will directly address. This creates cognitive tension that motivates engagement with the evidence-gathering phases.	×8 vs. ×2 for part (c) — if fewer than 30% write ×8, the teacher knows to slow down during Phase 3 and re-derive V.S.F explicitly before the application problems.
Phase 2 — Observe (20 minutes) 📹 **VIDEO:** [Geometry and motion in Borromini's San Carlo] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in their established groups of 4–5. Each group receives a differentiated problem card (three card types, rotated so each table tackles a different context). Card A: Survey of Kenya map — a 1 : 50 000 topographic map extract showing the road between Kisumu and Ahero; learners measure two distances on the map and calculate real distances, then determine the real area of a rectangular rice paddy shown on the map. Card B: Architectural plan — using the school plan at Scale 1 : 200, learners calculate the real floor area of the staffroom, the real volume of a square water tank shown on the plan with a given plan height, and determine how many 20-litre jerricans the tank can fill. Card C: Packaging design — a mandazi factory in Eldoret	Distribute problem cards and say: 'Your job in the next 15 minutes is to solve every part of your card using the L.S.F, A.S.F, and V.S.F relationships we have been building since Lesson 1. Show every formula before substituting numbers.' Circulate every 2–3 minutes. At each group, crouch to eye level and ask one targeted question rather than explaining: 'What is your L.S.F here? Good — so what is your A.S.F?' or 'You have the plan area — have you squared or cubed the scale factor?' Use 10-second wait time after each question. If a group is stuck on Card B's volume, prompt: 'The plan shows a 3 cm × 3 cm base and 2 cm height. What are the REAL dimensions first?' After 15 minutes, alert groups: 'Two minutes to finalise your answer sheet.' At 17 minutes, initiate a	Contextualised problem-card jigsaw: Differentiating contexts (map / plan / packaging) ensures learners encounter L.S.F, A.S.F, and V.S.F in genuinely distinct real-world settings rather than abstract repetition. The jigsaw then requires each learner to explain their context to peers who did not solve it, deepening understanding through peer teaching and exposing the common underlying mathematics across all three contexts.	Observable indicator: On Card B, learners correctly write A.S.F = 200^2 = 40 000 and V.S.F = 200^3 = 8 000 000 before calculating. Teacher checks that units are converted correctly (cm^2 on plan → m^2 in reality). Red flag: any group that multiplies the real length × the real width using plan dimensions without applying the scale factor — teacher intervenes immediately with the prompt above.

	wants to scale up its box design by L.S.F = 1.5; learners calculate the new surface area of cardboard needed and the new volume of mandazi the box holds, given the original box dimensions. All groups record working and answers on a shared group answer sheet.	jigsaw: one member from each Card-A group moves to a Card-B group, and one Card-B member moves to a Card-C group, and one Card-C member moves to a Card-A group, so every new group now has one expert on each card type.		
Phase 3 — Explain (20 minutes) 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	After the jigsaw reshuffle, each expert member takes 2 minutes to walk their new group through their card's solution, explaining which scale factor they used and why. Groups then collaboratively answer two synthesis questions written on the board: (Q1) 'The two water tanks in our school phenomenon look the same shape but one is twice as tall. Using the correct V.S.F, confirm: does the taller tank hold 2×, 4×, or 8× as much water? Show working.' (Q2) 'A maize storage silo in Eldoret and a scale model of it have L.S.F = 1 : 30. The model's curved surface area is 94 cm². What is the real silo's curved surface area in m²? And if the model holds 0.15 litres of grain, what volume of grain in m³ does the real silo hold?' Groups write solutions on mini-whiteboards or large paper for gallery display.	After the jigsaw explanations (allow 6 minutes total), bring the class together and say: 'Everyone look at Synthesis Question 1. I want silence for 15 seconds — think about which scale factor applies to volume.' Wait the full 15 seconds, then cold-call: 'Kipchoge, what is your answer and how did you get it?' After response: 'Does anyone want to challenge or extend Kipchoge's reasoning?' Cold-call a second learner. Confirm: 'V.S.F = $k^3 = 2^3 = 8$. The taller tank holds 8 times as much — exactly as our phenomenon showed us on Day 1. This is not a coincidence. This is the mathematics.' For Q2, ask groups to hold up their whiteboards simultaneously on your count of three. Scan for correct A.S.F = $30^2 = 900$ and correct unit conversion ($94 \times 900 = 84\ 600$ cm² = 8.46 m²). Cold-call one group to explain their unit conversion step. Narrate: 'This is exactly what a Survey of Kenya cartographer does every time they produce a map — they are using A.S.F and V.S.F, whether they name it that or not.'	Simultaneous whiteboard reveal with targeted cold-calling: Requiring all groups to display answers simultaneously prevents anchoring on a single early response and allows the teacher to identify the spread of understanding instantly. Connecting correct answers back to the anchoring phenomenon (the two tanks) closes the cognitive loop opened in Phase 1 and reinforces that the unit's central question has now been answered with rigorous mathematics.	Observable indicator: At least 80% of groups correctly write V.S.F = 8 for Q1 with working shown. For Q2, teacher checks that learners apply A.S.F (not L.S.F) to the surface area and V.S.F (not A.S.F) to the volume — a common error is to cube both or square both. Any group making this error is flagged for targeted support during Phase 4.
Phase 4 — Driving Question Board (DQB) Update (10 minutes) 📹 **VIDEO:**	Each learner writes two sticky notes (or two lines in their DQB section of their exercise book): one GREEN note beginning 'I can now explain…' and one ORANGE note beginning 'I still wonder…'. Groups then nominate one representative	Say: 'Take 3 minutes — silent individual writing. Green note: what can you now explain that you could not explain in Lesson 1? Orange note: what is still puzzling you?' After 3 minutes, invite three volunteers to read their green	Traffic-light DQB annotation with new generative question: Revisiting earlier DQB questions allows learners to experience the progression of their own understanding as a visible, cumulative record — a key	Observable indicator: Green notes use specific mathematical language — 'V.S.F = k^3', 'A.S.F = k^2', 'scale factor', 'proportion'. Vague notes ('I understand scale now') indicate surface learning; teacher notes these learners for

Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	to physically move or annotate cards on the class DQB. The class reviews three DQB questions posted in earlier lessons — 'Why does doubling the size of a water tank more than double the water it holds?', 'How do architects make sure every measurement on a plan stays in proportion?', and 'Can we use these rules for any solid, not just cubes?' — and decides together whether each can now be marked ANSWERED, PARTIALLY ANSWERED, or STILL OPEN. The teacher adds one new DQB card: 'When we build our models, how will we check that our two solids are truly similar and not just roughly the same shape?'	notes aloud. For each, ask the class: 'Does anyone want to add to or refine what [name] said?' Use 10-second wait time. Then direct attention to the DQB: 'Let's look at the question about the water tank. On a count of three, show me thumbs up for ANSWERED, sideways for PARTIAL, down for STILL OPEN. One, two, three.' After the class vote, ask: 'Wanjiku, you showed thumbs up — convince us in one sentence.' Add the new card to the DQB and say: 'This new question will be answered by your models next lesson. Keep it in mind as you plan today.'	metacognitive tool. Adding a forward-pointing question that connects directly to the project maintains intellectual momentum and primes learners for the model-building phase.	one-on-one check during Phase 5. The class vote on 'ANSWERED' questions gives real-time class-level comprehension data.
Phase 5 — Model Building (20 minutes) 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The model-building project is formally launched. Each group (pre-assigned groups of 4–5) receives a Project Brief Card specifying: they must construct AT LEAST TWO similar solid models from the same category (cube, cuboid, cylinder, or sphere), using a chosen L.S.F of their own selection (minimum k = 1.5, maximum k = 3), and using locally available materials — cardboard, clay, bottle caps, dried maize cobs, or recycled plastic bottles. Today's task: (i) Select their solid type and L.S.F. (ii) Complete a Project Planning Sheet: choose original dimensions, calculate scaled dimensions using L.S.F, calculate A.S.F and expected surface area of Model 2, calculate V.S.F and expected volume of Model 2. (iii) Sketch a labelled net or 3D diagram of both models with all dimensions marked. (iv) List the materials they will bring from home or source on the school compound. Groups that finish	Distribute Project Brief Cards and Planning Sheets. Read aloud the brief in 90 seconds, then say: 'You have 18 minutes. The most important thing you do today is the calculation table on your Planning Sheet — because if your maths is wrong, your models will not be similar. Accuracy first, construction second.' Circulate continuously. At each group, ask: 'What L.S.F did you choose? What is your A.S.F? Show me the calculation.' Use 10-second wait time. Specifically visit the groups flagged during Phase 3 for A.S.F/V.S.F confusion and work through one example with them: 'If your cylinder has radius 3 cm and you chose k = 2, what is the radius of your bigger model? What is the ratio of their curved surface areas?' At the 15-minute mark, say: 'You have 3 minutes — every group must have a completed calculation table and at least one labelled sketch before we close.' In the final 2 minutes, ask each group to state their solid	Project-anchored model planning as cumulative sense-making: Requiring learners to calculate dimensions BEFORE constructing forces them to use L.S.F, A.S.F, and V.S.F as functional tools rather than abstract formulae. The physical models will become tangible, persistent evidence that the scale-factor relationships are real and accurate — directly answering the anchoring phenomenon's puzzling observation about the two water tanks.	Observable indicator: Each group's Planning Sheet shows a correctly completed calculation table with three rows — L.S.F, A.S.F (= L.S.F^2), V.S.F (= L.S.F^3) — all derived from their chosen k, plus correctly scaled dimensions for both models. Teacher collects all Planning Sheets at the end of the lesson to check calculations before Lesson 8 construction resumes. Any group with an error in A.S.F or V.S.F receives written feedback on the returned sheet before the next lesson.

	planning early may begin cutting or shaping their first model.	type, L.S.F, and calculated V.S.F to the class in a 10-second 'show and tell' around the room.		

D. TEACHER REFLECTION
After the lesson, consider: (1) During Phase 2, did the jigsaw reshuffle create genuine peer teaching, or did some 'expert' members struggle to explain their card? If so, consider providing a one-sentence 'expert summary' scaffold on each card for Lesson 8's continuation. (2) In Phase 3, how many groups confused A.S.F with V.S.F when answering Synthesis Q2? If more than two groups made this error, open Lesson 8 with a 5-minute whole-class worked example before resuming construction. (3) Review all collected Planning Sheets tonight: are the V.S.F calculations correct? Flag any group whose V.S.F does not equal the cube of their stated L.S.F — this must be corrected before physical construction in Lesson 8, otherwise the models will not be genuinely similar and the project's mathematical integrity will be compromised. (4) Did the anchoring phenomenon (the school plan at Scale 1 : 200 and the two water tanks) feel vivid and motivating today, or has it become routine? If the latter, consider bringing a physical tape measure to Lesson 8 and actually measuring a classroom wall to verify the plan's scale in real time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined a Survey of Kenya map extract, the school architectural plan (Scale 1 : 200), and a mandazi packaging design problem — each presenting a real Kenyan context where lengths, areas, and volumes are related by L.S.F, A.S.F, and V.S.F respectively.
What did I learn?	Learners confirmed that when a linear scale factor k is applied: area scales by k^2 and volume scales by k^3 — so the taller water tank ($k = 2$) holds $2^3 = 8$ times as much water, not 2 times, resolving the cognitive conflict from the anchoring phenomenon. They applied these relationships to map distances, plan areas, tank volumes, and packaging design.
How does this explain the phenomenon?	Learners used L.S.F, A.S.F, and V.S.F to solve multi-step real-life problems, updated the Driving Question Board to mark the core phenomenon questions as ANSWERED, and produced a fully calculated Project Planning Sheet — including labelled model sketches and a materials list — to guide construction of their own pair of similar solid models in Lesson 8.

LESSON 8 (80 minutes): Project Presentations, Model Exhibition & Unit Consolidation — Final Explanation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate the entire sub-strand by presenting project models, articulating the full relationships among L.S.F, A.S.F, and V.S.F, and constructing a final annotated diagram that answers the driving question.
Knowledge	Learners demonstrate that when a solid is enlarged by a Linear Scale Factor k, the Area Scale Factor is k^2 and the Volume Scale Factor is k^3; they recall that scale drawings (e.g. 1 : 200) preserve all proportions while reducing or enlarging lengths only; and they relate these facts to the anchoring phenomenon of the school architectural plan and the two water tanks.
Skills	Learners present and explain physical models of similar solids; calculate and verify L.S.F, A.S.F, and V.S.F from their own constructed models; draw and annotate a cumulative object–image diagram labelling all three scale factors; and critically evaluate peers' models using mathematical reasoning.
Attitudes	Learners appreciate the use of similarity and enlargement in real-life situations (KICD verbatim); demonstrate integrity by using accurate measurements in their models; and show responsibility by caring for shared learning resources and the work of peers.
Key Inquiry Question	How are similarity and enlargement applied in day-to-day life?
Purpose in Storyline	This is the final lesson of the unit. Groups present the similar-solid models they have built over the unit (cubes, cuboids, cylinders, and spheres), state the L.S.F chosen, and verify the corresponding A.S.F and V.S.F. The class holds a mini-exhibition, then returns to the original anchoring phenomenon — the Kisumu school's 1 : 200 architectural plan and the two water tanks — to construct a complete, final verbal and diagrammatic explanation. The Driving Question Board is reviewed collaboratively, any residual questions are resolved, and each learner draws an individual cumulative annotated diagram as the summative consolidation activity.
Safety Notes	Ensure all model materials (card, wire, clay, scissors) are stored safely before the gallery walk. Learners should not run during the exhibition. Sharp tools used in model finishing must be placed flat on desks point-inward.

B. LESSON OVERVIEW
Lesson 8 is the capstone of the Similarity and Enlargement unit. It has three interlocking components: (1) a 20-minute Group Project Presentation and Mini-Exhibition in which each group displays their similar-solid models, states the L.S.F, and publicly demonstrates the squared and cubed relationships; (2) a 25-minute Driving Question Board (DQB) Review and Final Explanation in which the class collectively revisits the anchoring phenomenon — the Kisumu school plan (Scale 1 : 200) and the two water tanks — and articulates the complete answer to the driving question; and (3) a 25-minute Individual Cumulative Model-Building activity in which every learner draws and annotates a single object-to-image diagram labelling L.S.F, A.S.F, and V.S.F, writes a paragraph answering the driving question in their own words, and reflects on how their initial prediction (Lesson 1) compares with their final understanding. A 10-minute lesson close reviews outstanding DQB questions and previews Sub-Strand 2.2 (Reflection and Congruence).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 —	Learners individually re-read the	Display the original Lesson 1	Elicit-and-Compare: by juxtaposing	Teacher scans the two-sentence

Predict: Revisiting the Opening Prediction 📹 **VIDEO:** [Geometry and motion in Borromini's San Carlo] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	prediction they wrote in Lesson 1 about what happens to area and volume when a shape is scaled by factor 2. Without assistance, they write two sentences: (a) what they believed then, and (b) what they now believe, giving one piece of mathematical evidence from any previous lesson. They also silently examine the two water tank images displayed at the front (one tank appears twice as tall as the other) and predict — using precise mathematical language — the ratio of their volumes before the presentations begin.	phenomenon image (school plan + two water tanks) on the board. Say verbatim: 'Before we hear from any group, take 3 minutes in silence — look at your Lesson 1 prediction card and write two sentences: what you believed on Day 1, and what you now know to be true. Use a number or formula as your evidence.' Set a visible 3-minute timer. Circulate and note 3–4 learners whose Lesson 1 prediction was the common misconception ('double the size → double the volume') — you will cold-call these learners first in the debrief. After 3 minutes call on 2 learners who held the misconception: 'Amina, read us your Lesson 1 prediction, then tell us what you know now.' Provide 12-second wait time after each response. Affirm growth explicitly: 'The class has moved from intuition to proof — that is exactly what mathematicians do.'	the learner's Lesson 1 intuitive prediction with their Lesson 8 evidence-based statement, the strategy makes conceptual growth visible and creates cognitive readiness to engage with the final explanation. This mirrors the NGSS practice of Engaging in Argument from Evidence.	written comparisons during circulation. Observable indicator: learner correctly identifies that volume scales as k^3 (not k) and cites a specific numerical example (e.g., 'L.S.F = 2, so V.S.F = 8 as we found with the cylinder in Lesson 6'). Learners who still write 'volume doubles' are flagged for targeted questioning during Phase 2.
Phase 2 — Observe: Group Project Presentations and Mini-Exhibition 📹 **VIDEO:** [Geometry and motion in Borromini's San Carlo] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Similarity (Flexbook)] Source: CK-12 🔍 Search ARES	Each group (5–6 groups of 5–6 learners) has 3 minutes to present their pair of similar solid models at their desk station. During the presentation the group must: (i) state the solid type (cube, cuboid, cylinder, or sphere) and the L.S.F chosen; (ii) show one measurement on each model and demonstrate that the ratio equals the L.S.F; (iii) calculate the A.S.F = k^2 and V.S.F = k^3 live and point to where this is visible on the model; and (iv) state one real-life Kenyan application of their chosen scale factor (e.g., 'A jiko manufacturer in Kamukunji who doubles the radius of a charcoal jiko increases its cooking surface area by 4 times and its volume by 8 times'). After all presentations, a 5-minute silent	Brief groups 2 minutes before Phase 2 begins: 'You have exactly 3 minutes. Show us the L.S.F, prove the A.S.F, prove the V.S.F — use your ruler on the model if needed.' Use a timer visible to all. Cold-call one peer evaluator per group after each presentation: 'Kipchoge, what is one thing this group proved correctly, and one thing you want to challenge or clarify?' Allow 10-second wait time. After all presentations announce: 'Gallery walk — 5 minutes of silence. Place your sticky notes. You are looking for evidence of k, k^2, and k^3.' After the gallery walk, select 3 sticky notes aloud that contain substantive mathematical observations and read them to the class: 'Listen — this note says: I	Structured Gallery Walk with Claim-Evidence Sticky Notes: learners act as peer reviewers, which activates the NGSS practice of Obtaining, Evaluating, and Communicating Information. Requiring mathematical evidence (k, k^2, k^3) on each sticky note prevents superficial observation and forces learners to apply the core relationships to novel models they did not build.	Observable indicators: (a) each group correctly states A.S.F = k^2 and V.S.F = k^3 during presentation (teacher records on a 6-row checklist: group name, L.S.F stated, A.S.F correct Y/N, V.S.F correct Y/N, Kenyan application given Y/N); (b) gallery-walk sticky notes contain numerical references to scale factors rather than only aesthetic comments. Groups where A.S.F or V.S.F was incorrect are noted for targeted correction in Phase 3.

for similar readings	gallery walk allows every learner to visit at least two other groups' stations and write one 'I notice…' and one 'I wonder…' observation on a sticky note placed on each model.	wonder why the V.S.F for the sphere group is 27 but for the cuboid group it is also 27 even though their shapes are different. Does anyone want to answer that right now?' Provide 15-second wait time, then cold-call.		
Phase 3 — Explain: DQB Review and Final Explanation of the Phenomenon 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to seats. The teacher uncovers the full Driving Question Board that has been built across Lessons 1–7. Learners spend 2 minutes silently reading all posted questions and claims. The teacher then facilitates a whole-class discussion structured around three prompts projected on the board: (1) 'The Kisumu school plan is Scale 1 : 200. A classroom that is 10 m wide appears 5 cm on the plan. Why is only the length divided by 200 — not the area or the volume?' (2) 'One water tank is twice as tall as the other and has the same shape. A plumber says the larger tank holds 8 times as much water. Is the plumber correct? Prove it using L.S.F, A.S.F, and V.S.F.' (3) 'If a Nairobi architect enlarges a building plan from Scale 1 : 500 to an actual building, what is the ratio of the actual floor area to the plan floor area?' Learners write answers individually for 5 minutes, then share with a shoulder partner for 2 minutes, then the teacher cold-calls 3 pairs to present to the class.	Uncover the DQB dramatically: 'This board has held our questions since Lesson 1. Let us answer them together today.' Read each of the three prompts aloud slowly. Say: 'You have 5 minutes — write your full answer with mathematics. This is your final explanation of the phenomenon that puzzled us on Day 1.' Circulate: target the 3–4 learners flagged in Phase 1 for still-incomplete understanding; crouch to their level and ask quietly: 'What is k for the water tank? Now what is k^3?' After partner share, cold-call Pair 1 for Prompt 1, Pair 2 for Prompt 2, Pair 3 for Prompt 3. After each pair presents, ask the class: 'Do you agree? Disagree? Add to this explanation?' Provide 12-second wait time. Explicitly close the DQB: cross out or tick each original question card as it is answered. Say: 'Notice — we have now answered every question on this board. That is the journey of a mathematician: from puzzlement to proof.' Correct any group from Phase 2 whose A.S.F or V.S.F was wrong by referencing their model: 'Group 3 — you said your A.S.F was 4. Let us confirm: L.S.F = 2, A.S.F = 2^2 = 4. Correct. V.S.F = 2^3 = 8. Well done.'	Three-Prompt Collaborative Final Explanation (Think–Pair–Share variant): the three prompts are deliberately sequenced — Prompt 1 tests L.S.F only, Prompt 2 integrates all three scale factors in context, Prompt 3 reverses the direction of reasoning (from scale factor to real-world ratio). This graduated release, grounded in the anchoring phenomenon, embeds the NGSS practice of Constructing Explanations and aligns with the KICD learning outcome: 'relate linear scale factor, area scale factor, and volume scale factor in enlargements.'	Observable indicators: (a) written answers for Prompt 2 correctly show L.S.F = 2, A.S.F = 4, V.S.F = 8 with conclusion 'the plumber is correct'; (b) Prompt 3 answers correctly compute $(500)^2$ = 250,000 as the ratio of actual floor area to plan floor area; (c) DQB tick-off is complete with no unanswered question cards remaining. Learners who cannot complete Prompt 3 independently are noted for follow-up in the next sub-strand's introductory lesson.
Phase 4 — Driving Question Board	Learners collaborate in groups of three to write a single 'Final Answer Card' (index card provided) that responds to the full	Distribute index cards. Say: 'In groups of three, you have 5 minutes to write the Final Answer Card for our driving question. Your	Final Answer Card (Collaborative Claim Construction): requiring learners to synthesise all three scale factor rules plus a novel	Observable indicators: (a) every Final Answer Card includes the symbolic statements A.S.F = k^2 and V.S.F = k^3; (b) Kenyan

Completion and Closure 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	driving question: 'When you scale up a shape or solid by a linear scale factor k, lengths scale by k, areas scale by k^2, and volumes scale by k^3. This is why the 1 : 200 school plan fits on an A3 sheet with all proportions correct, and why a tank that is twice as tall holds 8 times as much water.' Each card must include: (a) the three scale factor rules in symbolic form; (b) one Kenyan real-life example that is different from the water tank and the school plan; and (c) the group members' names. Cards are read aloud by one representative per group, then physically pinned to the DQB under the heading 'OUR FINAL ANSWER.' The teacher ceremonially removes the original driving question card, holds it up, and pins the stack of Final Answer Cards in its place.	example must be Kenyan — think of silos in Eldoret, matatu body fabrication in Nairobi, fish-smoking drums in Kisumu, water pots in Siaya — anything real.' Circulate and redirect groups whose Kenyan examples are vague ('a building') toward specificity ('the new Kisumu Sports Complex, whose swimming pool is scaled up from a training pool'). After 5 minutes, call one representative per group to read their card (30 seconds each). After all cards are pinned, pause for 8 seconds of whole-class silence looking at the completed DQB, then say: 'From confusion to clarity — that is what mathematics does for us and for our communities.' Preview Sub-Strand 2.2: 'Next lesson we begin Reflection and Congruence — you will find that congruence is actually the special case of similarity where k = 1.'	Kenyan example into a single concise card activates the NGSS practice of Constructing Explanations and Using Mathematics. The ceremonial DQB closure provides affective closure and reinforces that the inquiry arc (from phenomenon to explanation) is a replicable scientific and mathematical habit of mind.	examples are specific and mathematically plausible (teacher checks that the scale factor chosen is stated and the resulting A.S.F and V.S.F are correct on the card); (c) no group's card retains the Lesson 1 misconception that 'doubling size doubles volume.' Cards with errors are returned for a 2-minute revision before pinning.
Phase 5 — Individual Cumulative Model Building: Final Annotated Diagram 📹 **VIDEO:** Geometry and motion in Borromini's San Carlo Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Similarity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually draws a final cumulative annotated diagram in their exercise book. The diagram shows a clearly labelled object solid (a cylinder representing the smaller Kisumu school water tank, with radius r and height h) and its enlarged image (radius kr and height kh, where k = 2 is used as the concrete example). The learner annotates: (1) the L.S.F = k = 2 on a length arrow; (2) the A.S.F = k^2 = 4 on a curved surface area bracket; (3) the V.S.F = k^3 = 8 on a volume label inside the image; (4) a note connecting the diagram back to the phenomenon: 'This is why the large tank at our school holds 8× the water of the small tank, even though it is only 2× as tall.' Below the diagram, learners write a 3–5 sentence paragraph	Announce: 'For the final 20 minutes, every learner works individually and in silence. Draw the diagram, label all three scale factors, and write your paragraph. This is your personal record of everything we have learned.' Write the three required annotations on the board as a checklist: '✓ L.S.F = k labelled on a length; ✓ A.S.F = k^2 labelled on an area; ✓ V.S.F = k^3 labelled on a volume; ✓ Phenomenon connection sentence; ✓ Driving question paragraph; ✓ I used to think / Now I know.' Circulate continuously. At the 10-minute mark, cold-call 2 learners (choose one high-attainer and one who was flagged earlier) to read their 'I used to think / Now I know' aloud without showing their book — this models intellectual honesty. With 3 minutes remaining	Individual Cumulative Annotated Diagram with 'I Used to Think / Now I Know' Reflection: this strategy, drawn from Project Zero's Thinking Routines, makes the learner's conceptual trajectory visible to both learner and teacher. The requirement to connect every annotation back to the anchoring phenomenon ensures that abstract symbolic knowledge (k, k^2, k^3) is anchored in the concrete Kisumu context throughout. This embeds the NGSS practice of Developing and Using Models at its highest level — the learner produces the model independently, without scaffolding.	Observable indicators: (a) diagram correctly shows both object and enlarged image with all three scale factor annotations in the right locations (length, area, volume); (b) paragraph correctly states the three rules and applies them to at least one of the two phenomenon elements (school plan or water tanks); (c) 'I used to think / Now I know' entry shows a clear and mathematically accurate contrast between the Lesson 1 misconception and the correct k / k^2 / k^3 relationships. Teacher collects exercise books at the end of the lesson for holistic marking against the KICD assessment rubric (Exceeds / Meets / Approaches / Below Expectations) on the indicators: 'Ability to determine linear, area and volume scale factors' and 'Ability to apply

	answering the driving question in their own words, referencing the school plan and the water tanks. Finally, they revisit their Lesson 1 prediction card one more time and write: 'I used to think… Now I know… The evidence that changed my mind was…'	say: 'Finish your final sentence. You should be able to describe, in one sentence, what happens to length, area, and volume when you scale by k.' With 1 minute remaining: 'Pens down. Look at what you have drawn. This diagram — and your understanding — did not exist eight lessons ago.'		similarity and enlargement to real-life situations.'

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did every group correctly state A.S.F = k^2 and V.S.F = k^3 during their presentation, or did any group confuse the squared and cubed relationships — if so, what misconception persisted and how will you address it in Sub-Strand 2.2? (2) Were the Final Answer Cards mathematically precise and Kenya-specific, or did groups produce generic statements — what scaffolding would help future cohorts? (3) Did the 'I used to think / Now I know' entries in the individual diagrams reveal a genuine conceptual shift from the Lesson 1 prediction, or do some learners still conflate L.S.F with A.S.F — these learners need additional support before the sub-strand assessment? (4) Was 80 minutes sufficient for all five phases — if the mini-exhibition over-ran, consider reducing group presentation time from 3 to 2 minutes in future, or splitting the exhibition across Lessons 7 and 8. (5) Note the names of learners whose individual diagrams were collected and whose annotations were incomplete or incorrect; schedule a 10-minute catch-up session before the sub-strand written test. (6) Reflect on whether the DQB closure felt genuine or procedural — the ceremony of pinning Final Answer Cards over the original driving question is only powerful if learners have substantively engaged with all phases; use this reflection to judge the depth of understanding the unit has produced.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Groups presented physical models of similar solids (cubes, cuboids, cylinders, spheres) built from locally available materials; learners measured corresponding lengths on each model pair, computed ratios, and displayed their models during a silent gallery walk at desk stations around the classroom.
What did I learn?	When a solid is enlarged by a Linear Scale Factor k, the Area Scale Factor is k^2 and the Volume Scale Factor is k^3; these relationships explain why the Kisumu school's 1 : 200 architectural plan preserves all proportions while compressing every length by 200, and why a water tank that is only twice as tall holds exactly 8 times as much water as the smaller tank.
How does this explain the phenomenon?	Learners constructed individual cumulative annotated diagrams labelling L.S.F, A.S.F, and V.S.F on a cylinder object-image pair, wrote a driving-question paragraph connecting the three scale factor rules to the school plan and water tanks, completed Final Answer Cards with a novel Kenyan real-life example, and recorded a reflective 'I used to think / Now I know' entry documenting their conceptual journey from Lesson 1 intuition to Lesson 8 proof.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.